2026 Operating System

Linux on VM

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VirtualBox 설치

- VirtualBox란?
  - 오라클에서 개발한 오픈 소스 가상화 소프트웨어
  - 하나의 운영 체제(OS)에서 다른 운영 체제를 가상 환경으로 실행할 수 있음
    - 강의에서는 Guest OS로 Linux Ubuntu를 사용
  - Windows, macOS, Linux 등 다양한 운영 체제에서 사용 가능

1. Windows

1) VirtualBox 설치

- https://www.virtualbox.org

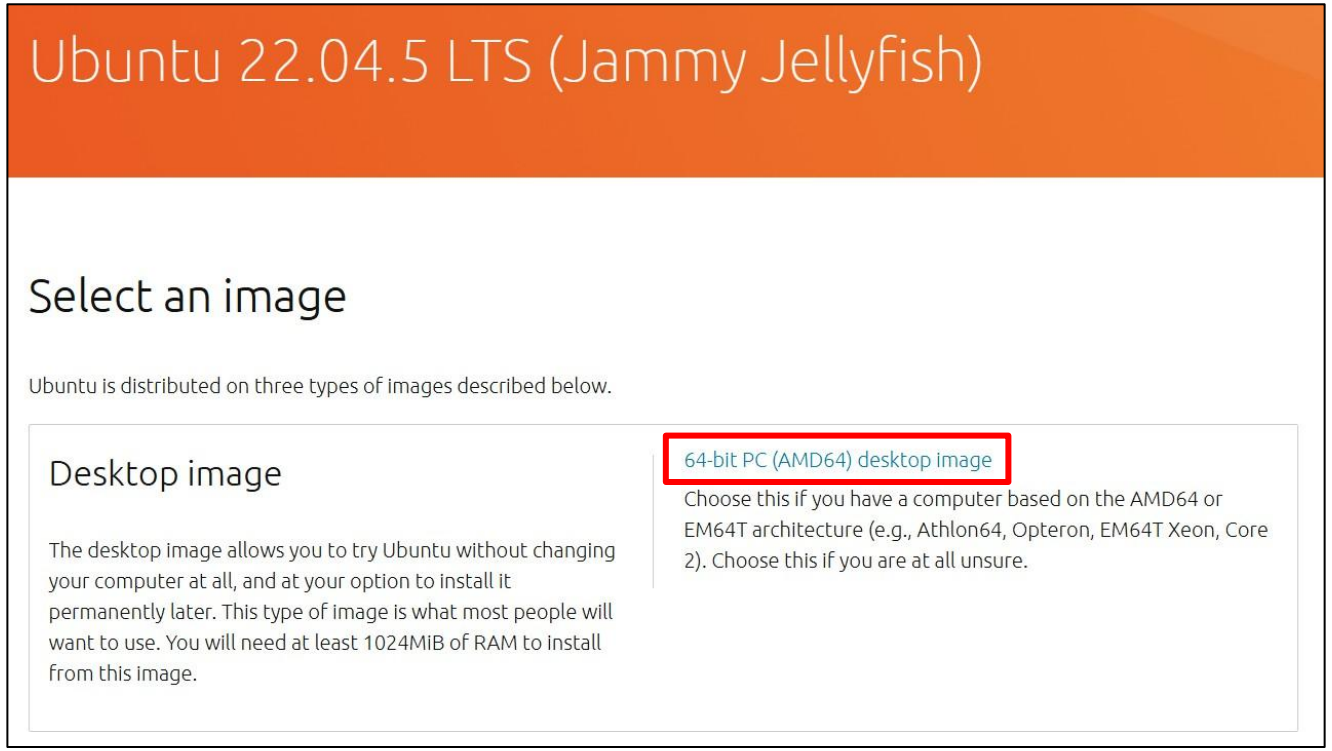
VirtualBox 다운로드 페이지

설치 후 VirtualBox 초기화면

1. Windows

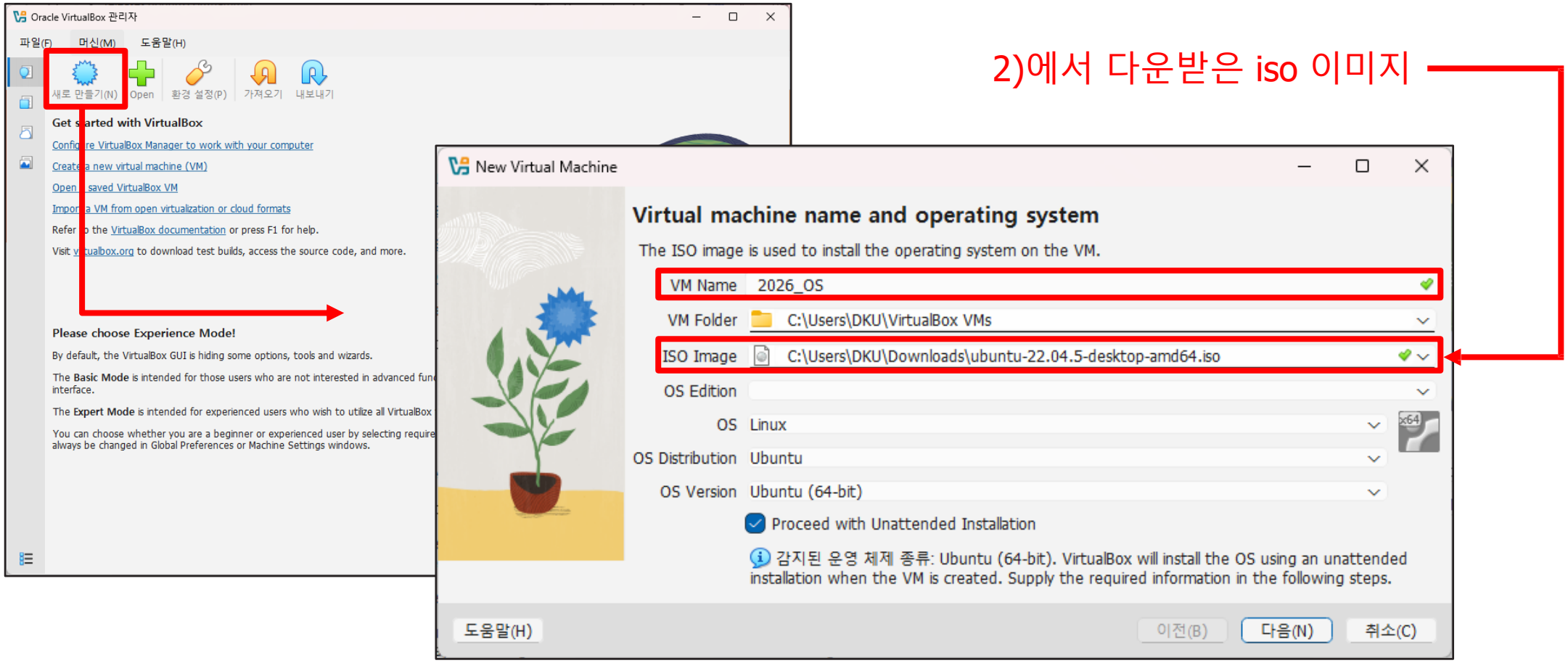
2) Ubuntu Desktop 22.04 이미지 다운로드 (AMD64)

- https://releases.ubuntu.com/jammy/



1. Windows

3) VM 생성 – VM 이름 및 ISO 이미지 선택

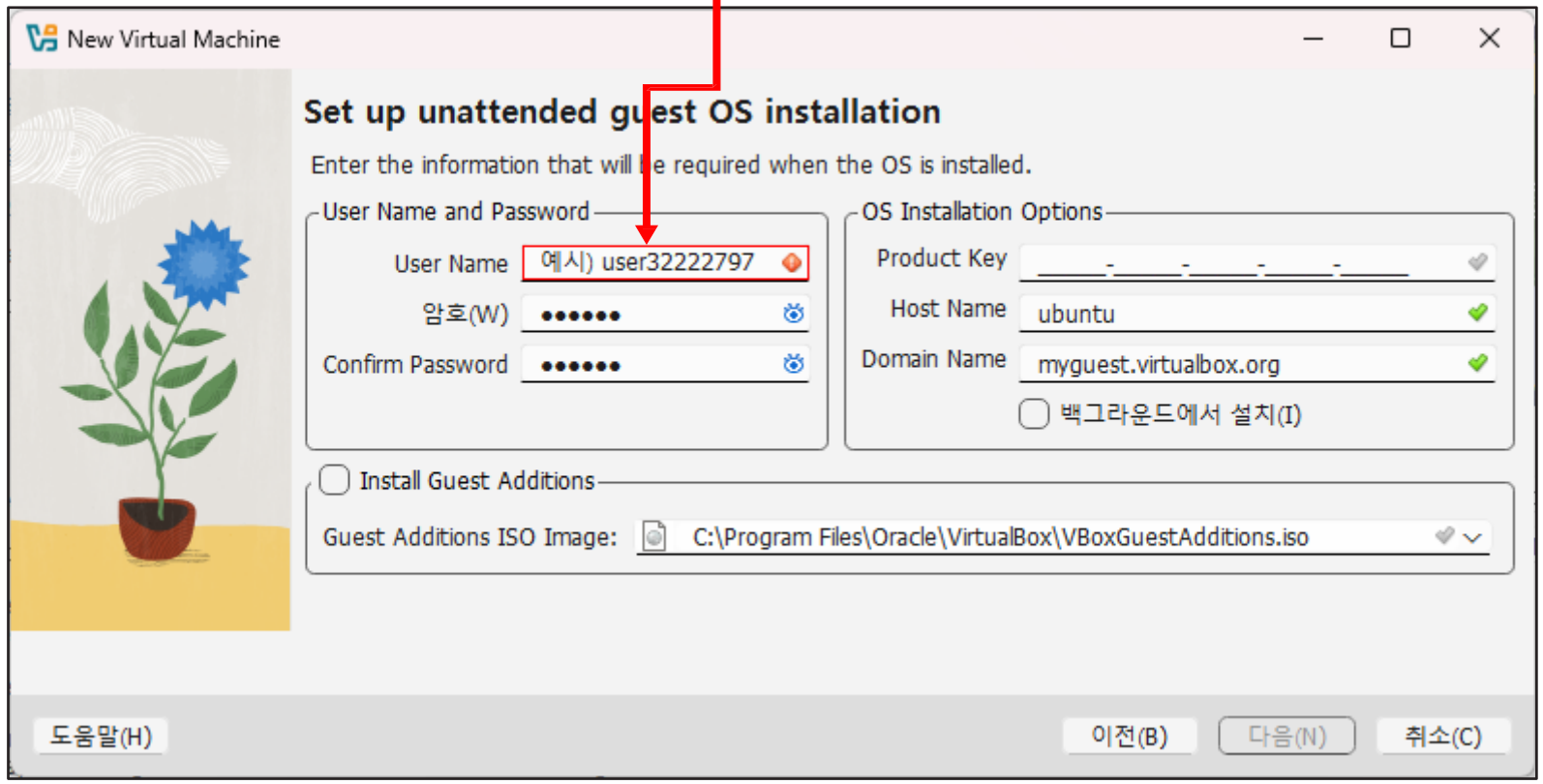


2)에서 다운받은 iso 이미지

1. Windows

3) VM 생성 –계정 생성

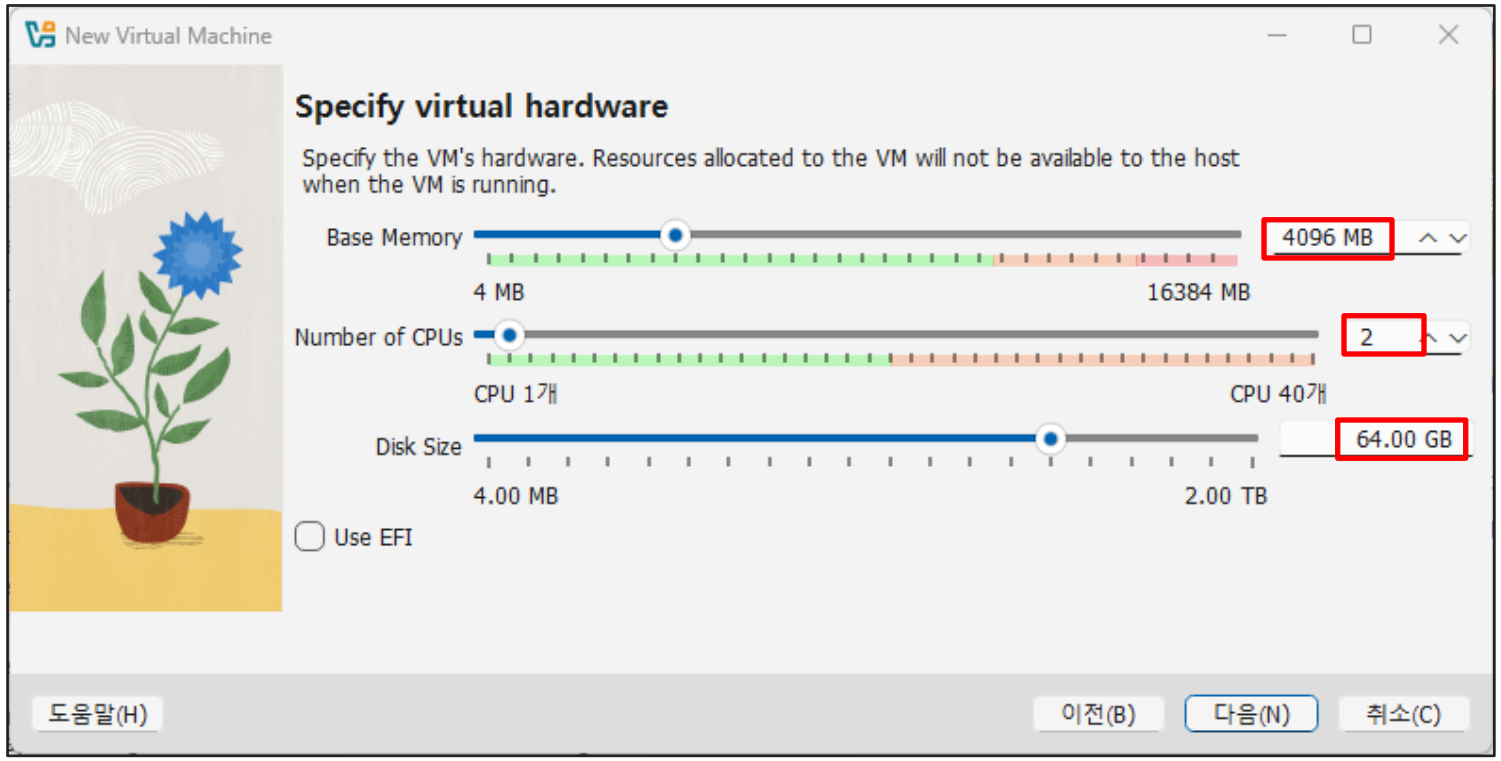
- 사용자 이름: **user + 학번으로 설정**



1. Windows

3) VM 생성 – 하드웨어 사양 선택

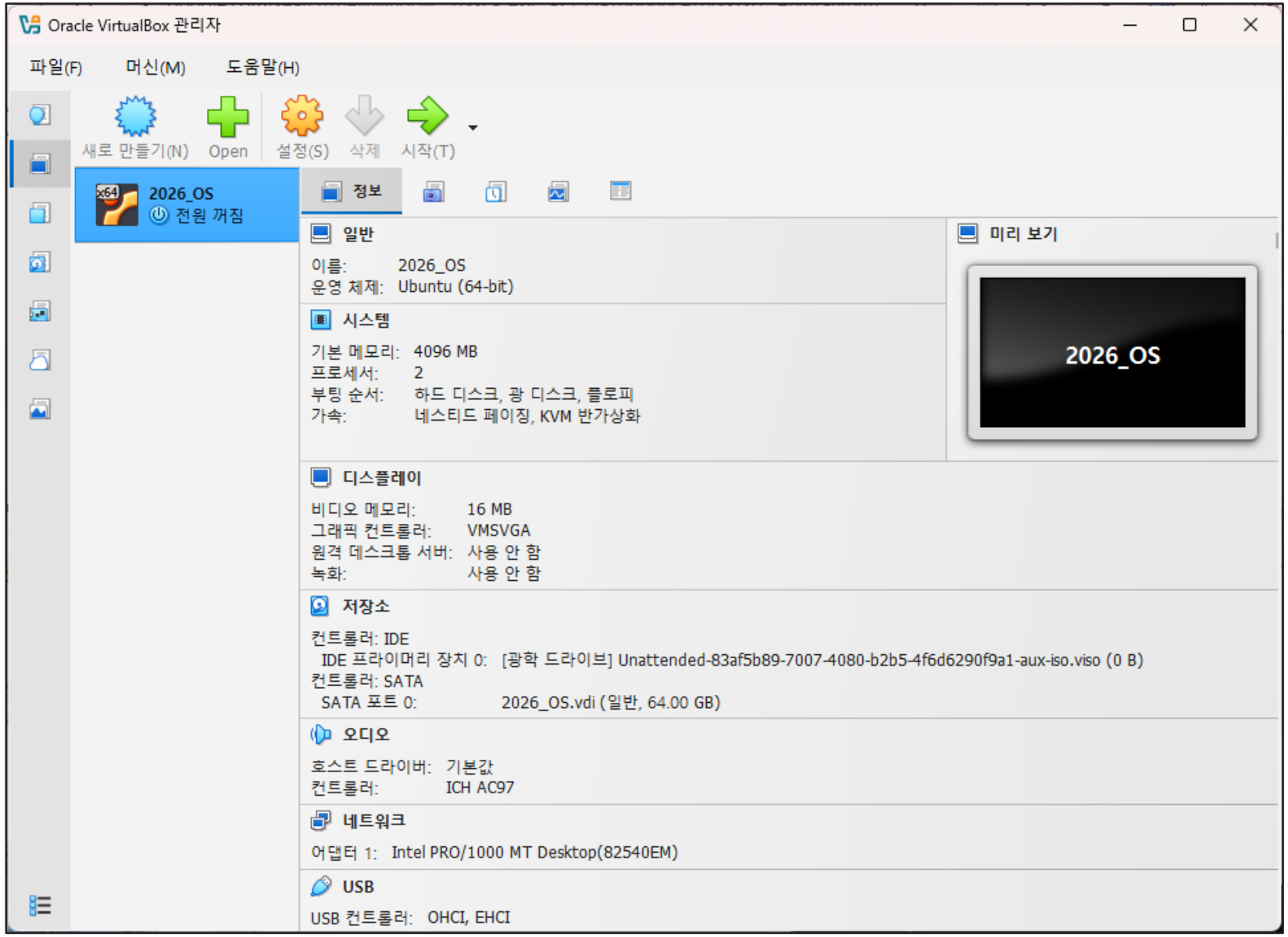
- 메모리: 4096MB, 프로세서: 2, 디스크: 64GB
- 본인 컴퓨터 사양에 맞춰 설정 가능



1. Windows

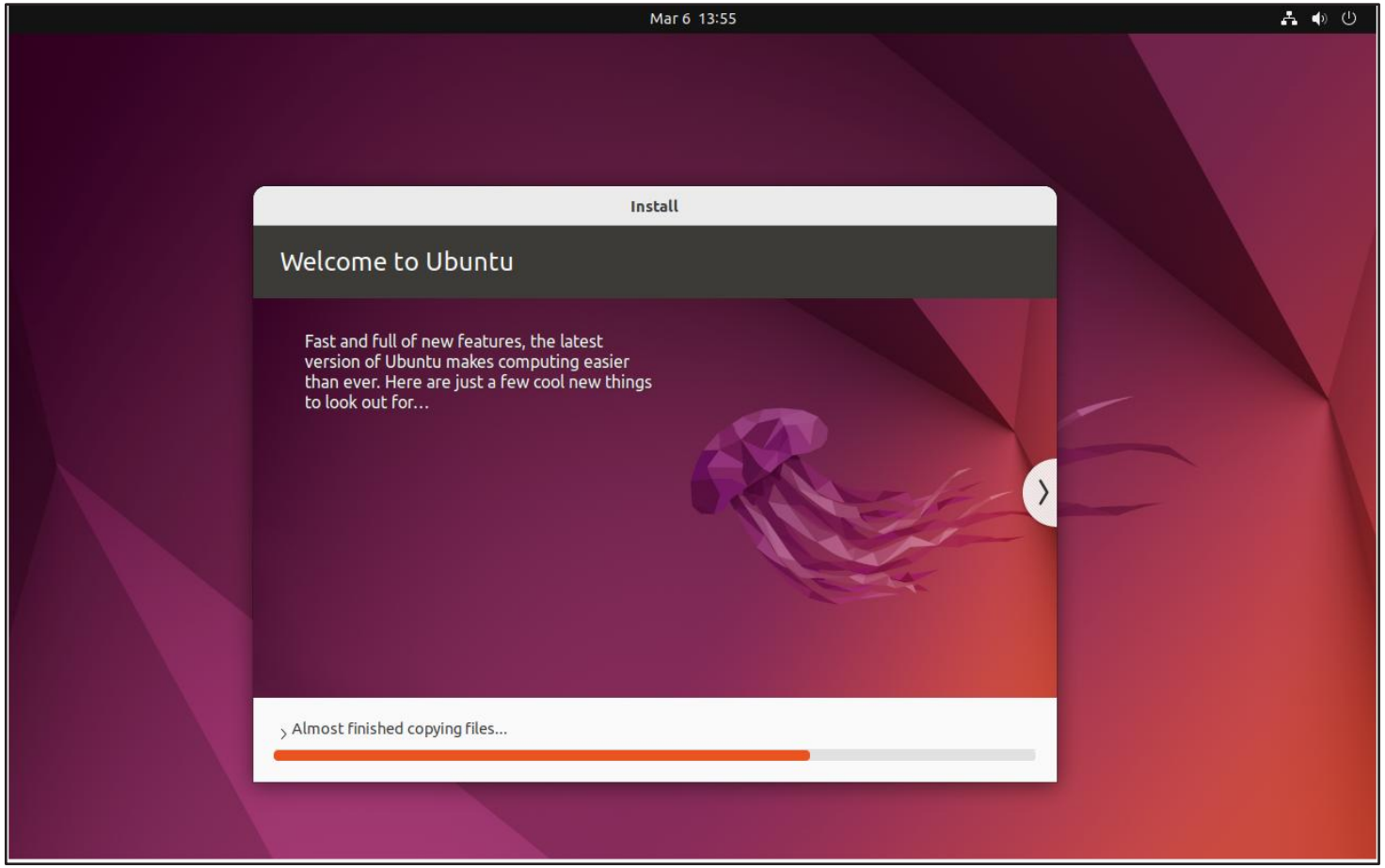
3) VM 생성 – 완료

- 더블 클릭해서 실행



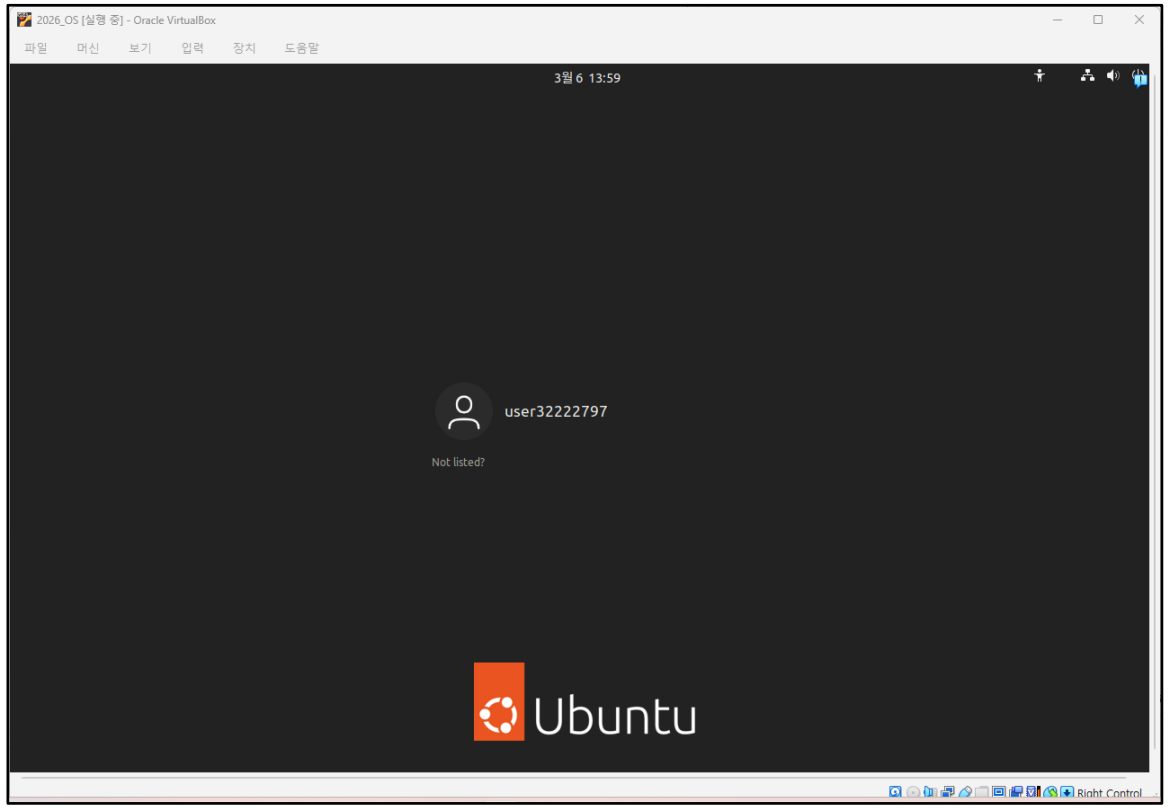
1. Windows

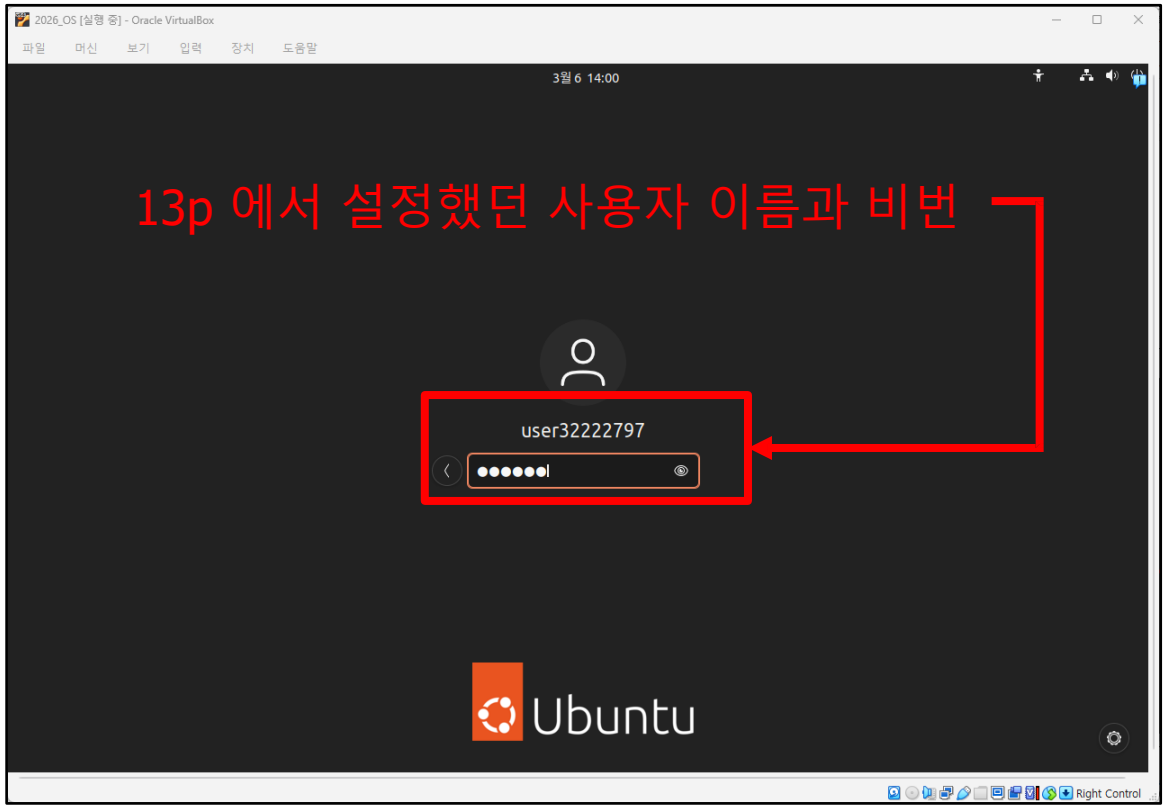
4) 초기 설정 – ubuntu 설치



1. Windows

4) 초기 설정 – 계정 확인 및 로그인

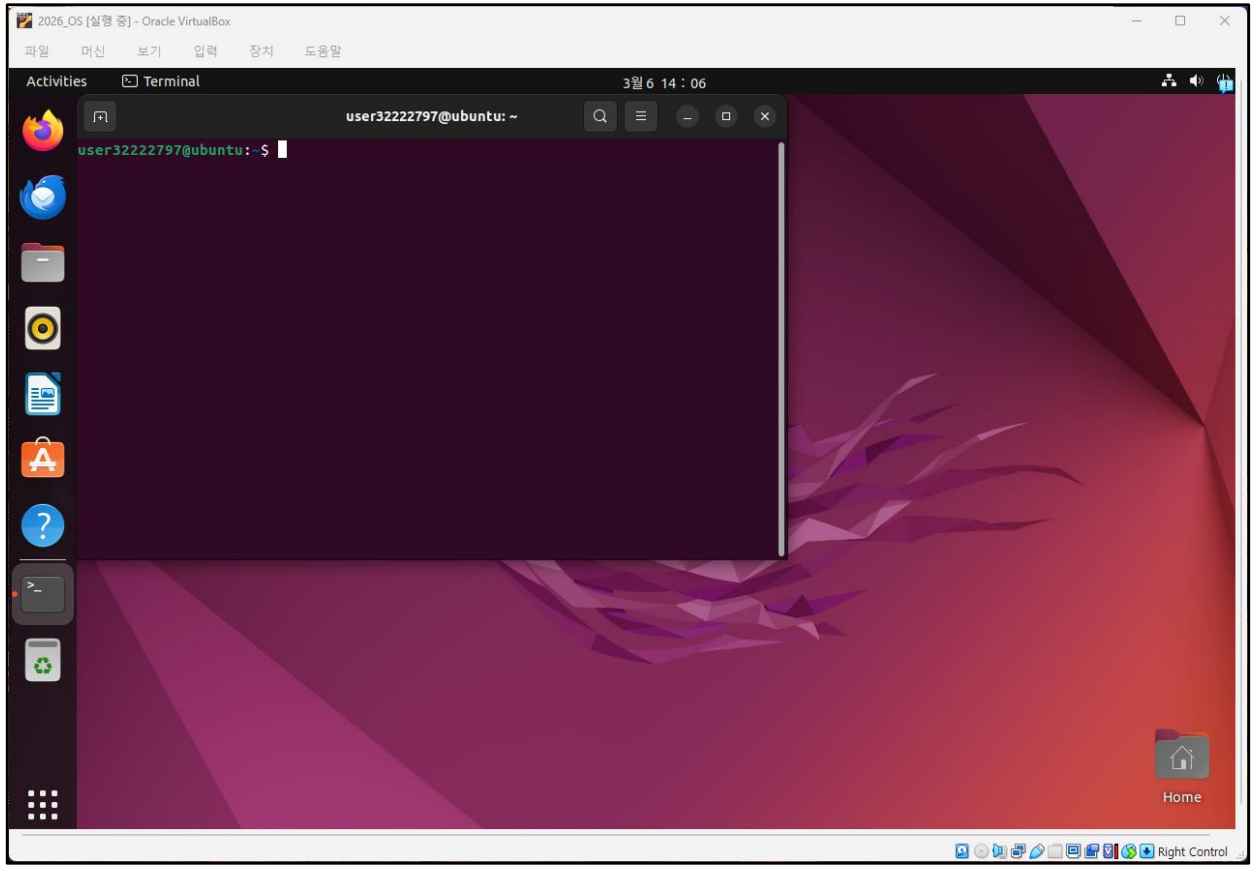




1. Windows

4) 초기 설정 – 터미널 열기

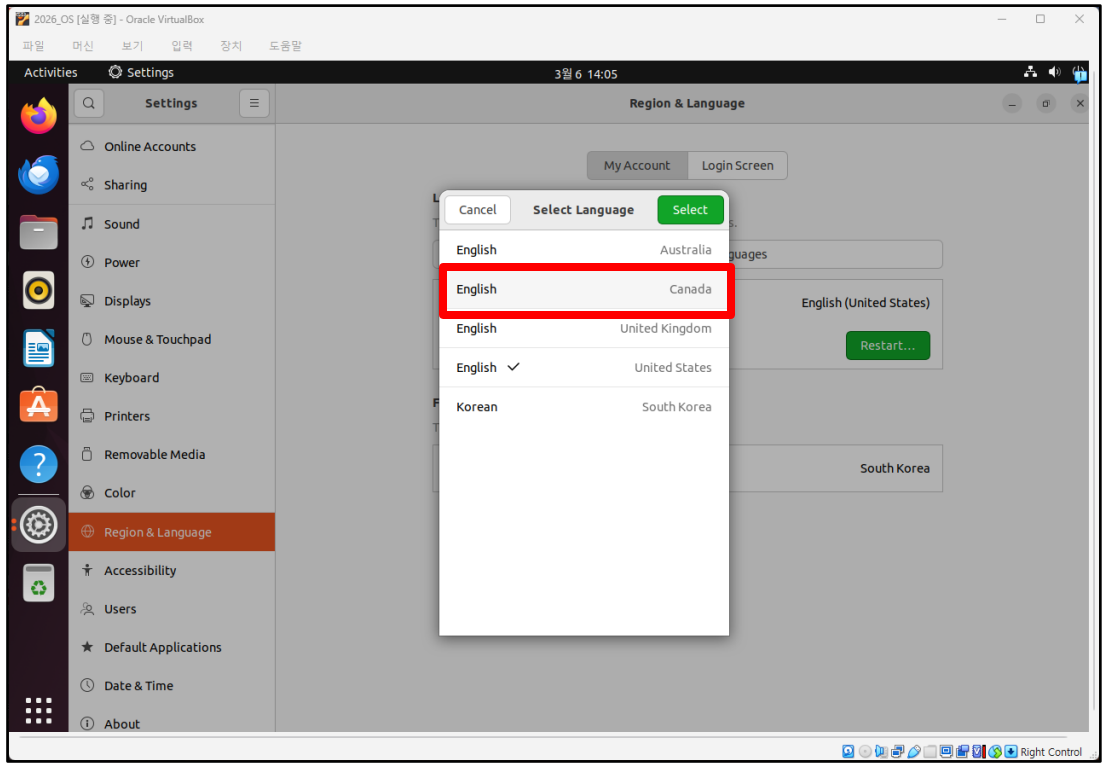
- 터미널 단축키: Ctrl + Alt + t



1. Windows

4) 초기 설정 – 터미널 열기

- 만약 터미널이 안 열린다면 Settings → Region & Language → Language
- English(United State)를 English(Canada)로 변경 후 restart



1. Windows

4) 초기 설정 – sudo 권한 설정

- sudo 명령어 실행 시 오류가 난다면 아래 과정 수행
- root로 전환 후 13p의 비밀번호 입력, {username}을 sudo 그룹에 추가하여 권한 부여
  - $ su
  - $ usermod –aG sudo {username}
  - $ reboot

```
root@ubuntu: /home/user32222797
user32222797@ubuntu:~$ su
Password:
root@ubuntu:/home/user32222797# usermod -aG sudo user32222797
root@ubuntu:/home/user32222797# reboot
```

- 재부팅후 sudo 명령어 작동 확인

```
user32222797@ubuntu: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

user32222797@ubuntu:~$ sudo apt update
[sudo] password for user32222797:
Hit:1 http://kr.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://kr.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://kr.archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:4 http://kr.archive.ubuntu.com/ubuntu jammy/main Translation-en_CA [4,224 B]
Get:5 http://kr.archive.ubuntu.com/ubuntu jammy/universe Translation-en_CA [1,13
2 B]
Hit:6 http://security.ubuntu.com/ubuntu jammy-security InRelease
Fetched 5,356 B in 1s (8,776 B/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
337 packages can be upgraded. Run 'apt list --upgradable' to see them.
user32222797@ubuntu:~$
```

1. Windows

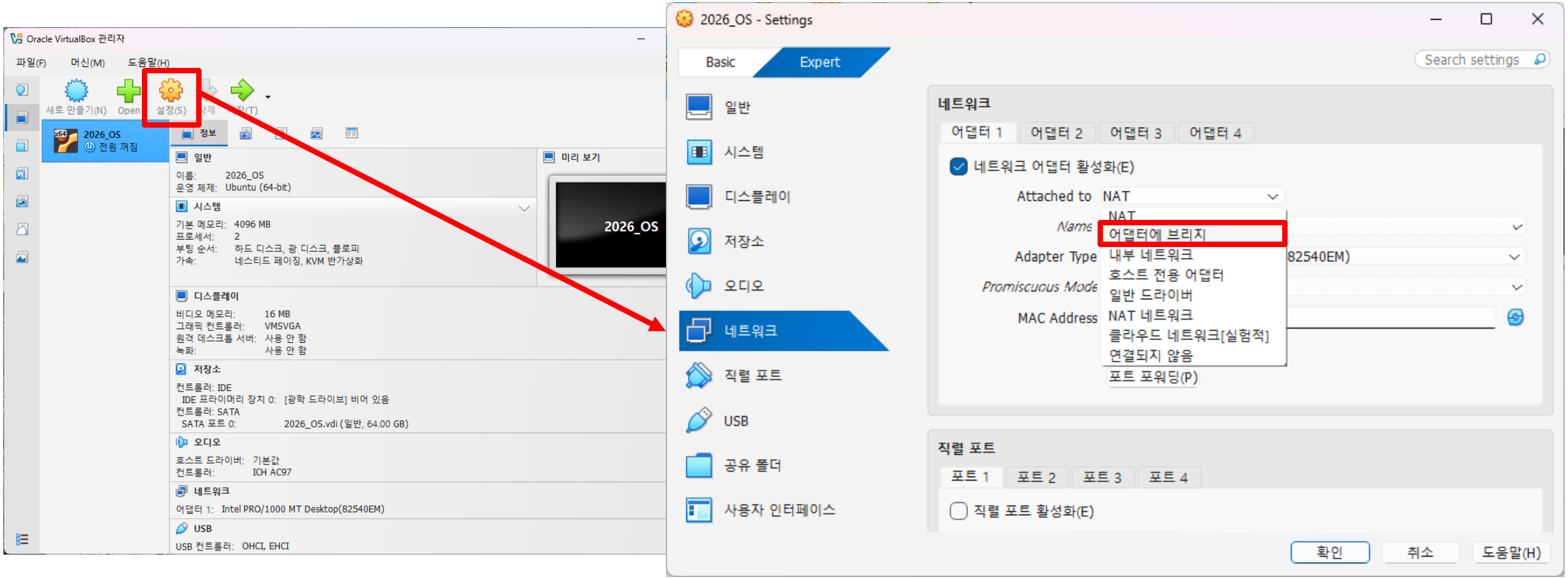
5) Optional – SSH 설정

- SSH: 다른 컴퓨터에 원격으로 접속하여 명령을 실행하고 정보를 주고받을 수 있는 프로토콜
- 호스트(윈도우)에서 VM(우분투)에 원격으로 접속해 터미널 사용가능

1. Windows

5) Optional – SSH 설정

- VirtualBox 네트워크 → 어댑터에 브릿지 설정



1. Windows

5) Optional – SSH 설정

- VM 실행 후 아래 명령어 실행
- $ sudo apt install –y openssh-server net-tools
  - openssh-server: SSH 접속을 가능하게 해주는 패키지
  - net-tools: ip 정보 및 네트워크 관련 정보를 확인할 수 있는 패키지
- $ systemctl status sshd
  - SSH 서버가 정상적으로 동작 중임을 확인

```
user32222797@ubuntu:~$ systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: e
     Active: active (running) since Fri 2026-03-06 14:35:01 KST; 10s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
```

- $ ifconfig
  - IP 주소 확인

```
user32222797@ubuntu:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.23.14.210  netmask 255.255.254.0  broadcast 172.23.15.255
        inet6 fe80::ed:40c2:abf8:eabe  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:43:0d:fe  txqueuelen 1000  (Ethernet)
```

1. Windows

5) Optional – SSH 설정

- Host(윈도우)에서 터미널 실행 (VM은 실행 중인 상태)
- > ssh {username}@{IP address}
  - IP address: 24p에서 확인한 IP (VM 재시작시 바뀔 수 있으니 확인 필요)

```
C:\Users\DKU>ssh user32222797@172.23.14.210
The authenticity of host '172.23.14.210 (172.23.14.210)' can't be established.
ED25519 key fingerprint is SHA256:prvBhbt7Qtg1c6y0U7MfjKYfxvx1wdvaWVp5j4ZxMrE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.23.14.210' (ED25519) to the list of known hosts.
user32222797@172.23.14.210's password:
```

입력되는 값이 뜨진 않지만
정상적으로 입력되고 있으니 비밀번호 입력 후 엔터

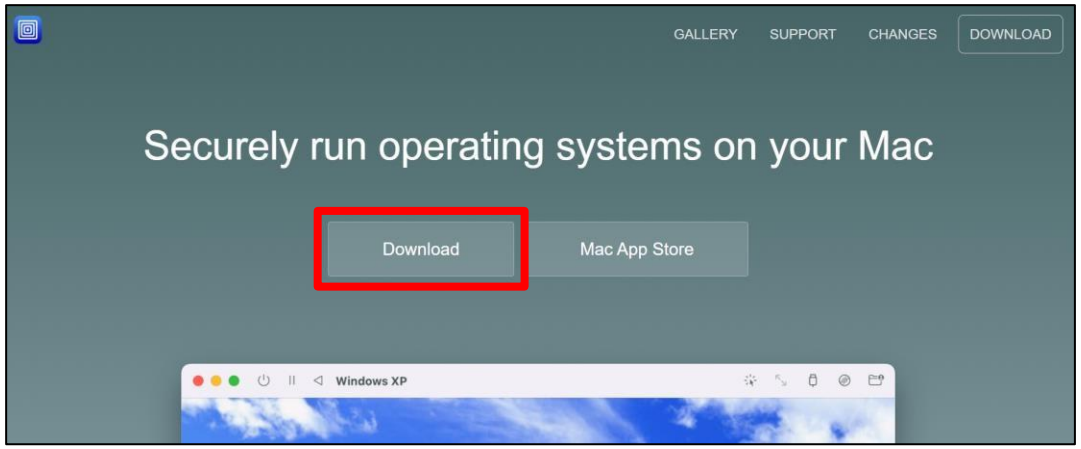
UTM 설치

- UTM이란?
  - macOS에서 가상화를 지원하는 오픈 소스 가상화 소프트웨어
  - Apple Silicon에서 다양한 운영 체제를 실행할 수 있음
  - Windows, Linux 등 다양한 운영 체제를 가상 환경에서 실행 가능
    - 강의에서는 Guest OS로 Linux를 사용

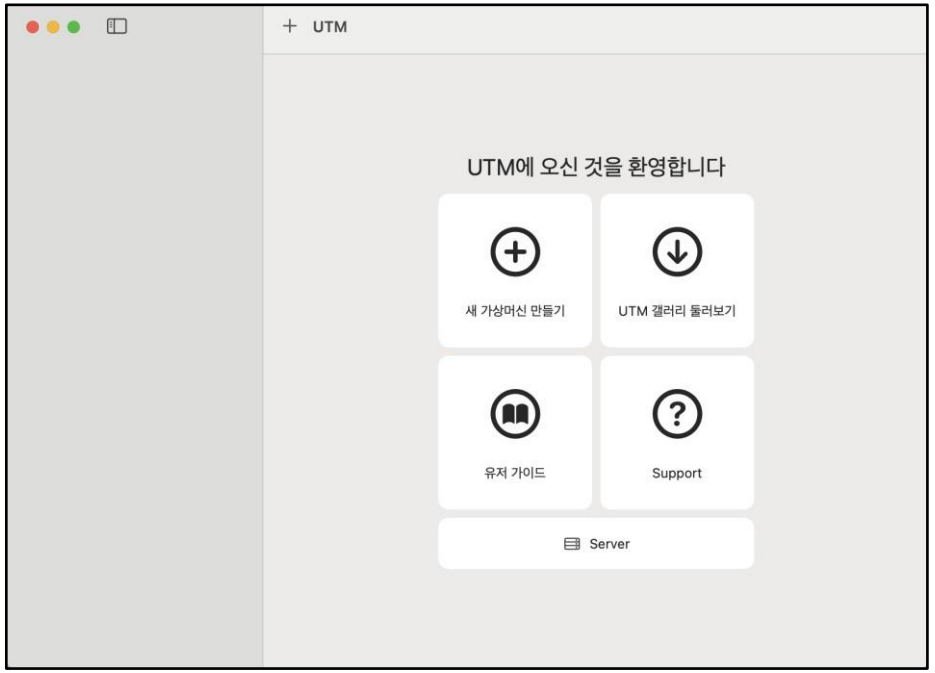
2. Mac (ARM)

1) UTM 설치

- https://mac.getutm.app/



UTM 다운로드 페이지

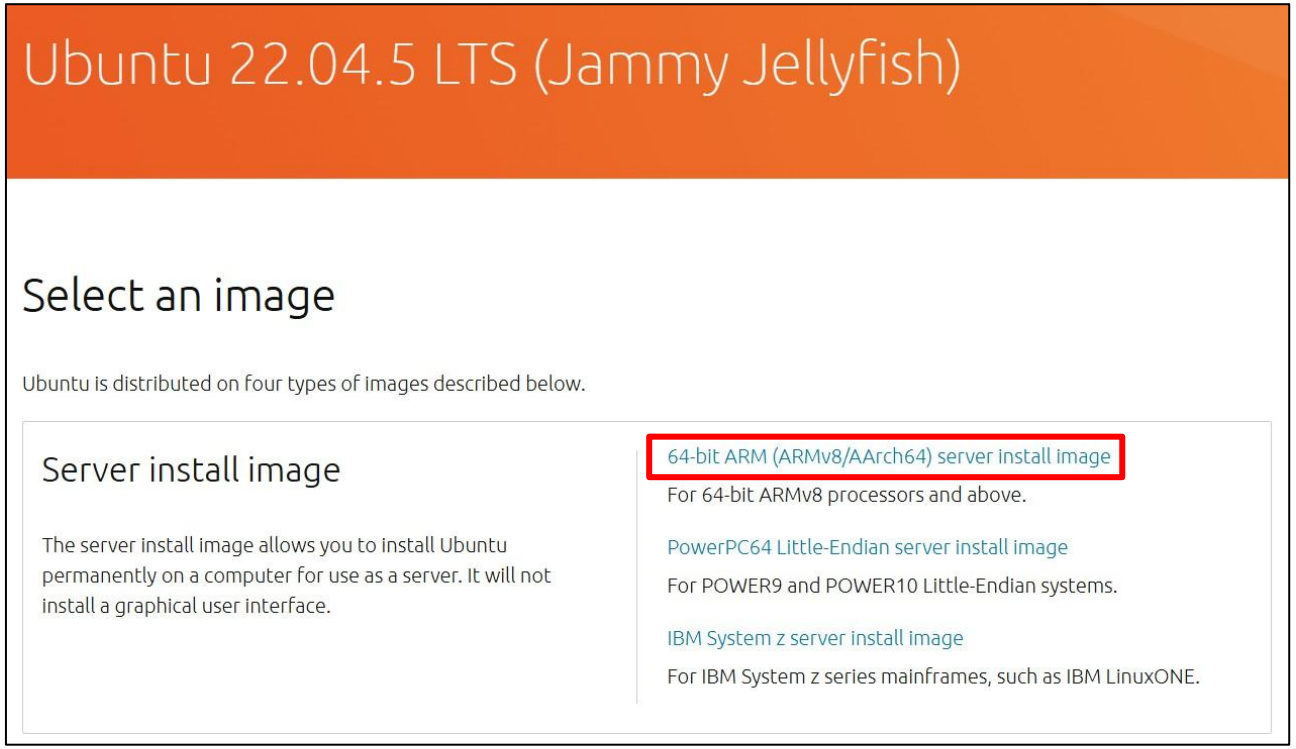


설치 후 UTM 초기화면

2. Mac (ARM)

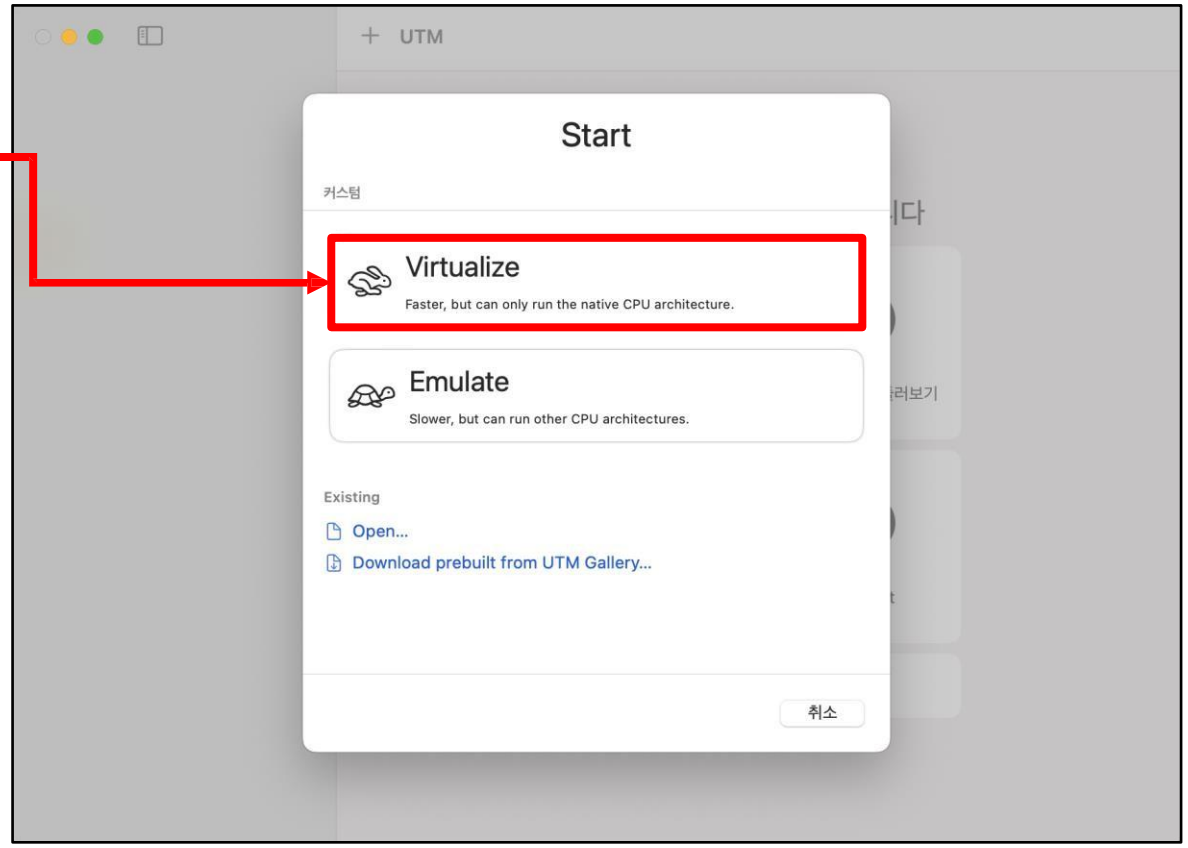
2) Ubuntu Desktop 22.04 이미지 다운로드 (ARM)

- https://cdimage.ubuntu.com/releases/jammy/release/



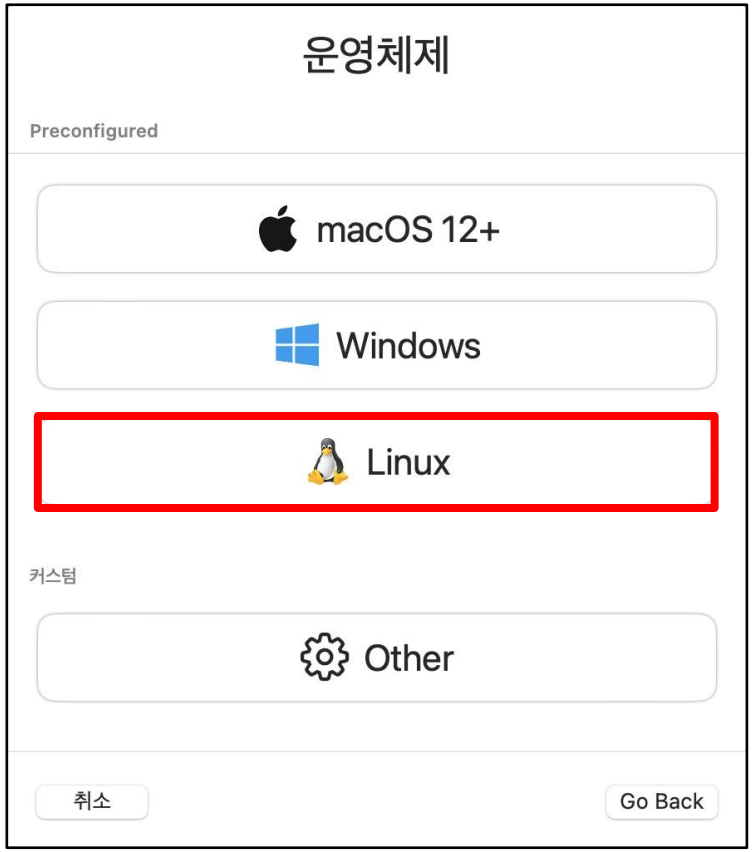
2. Mac (ARM)

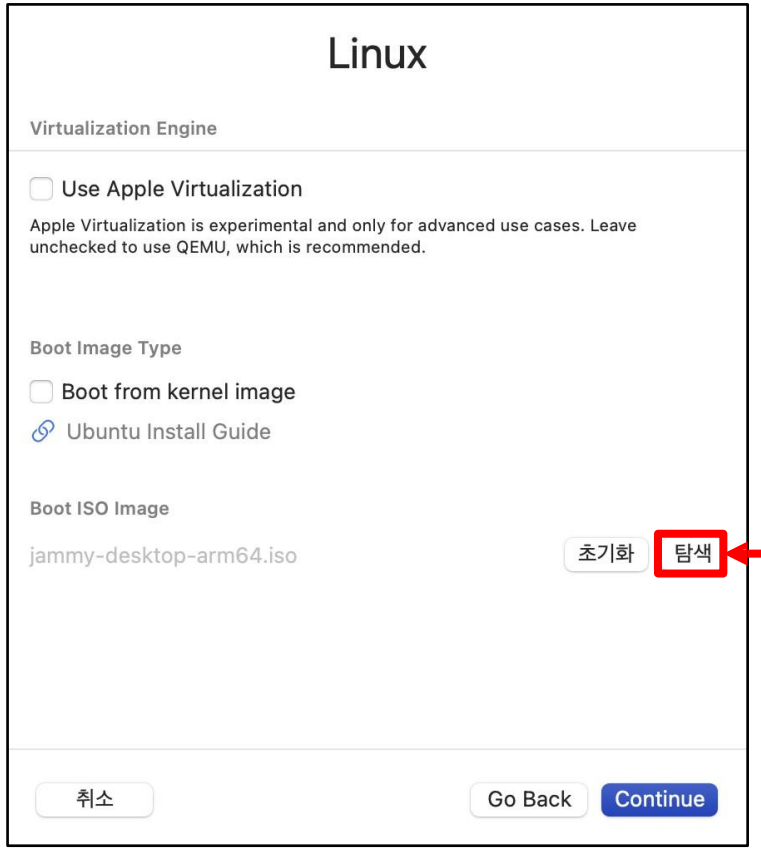
3) VM 생성



2. Mac (ARM)

3) VM 생성





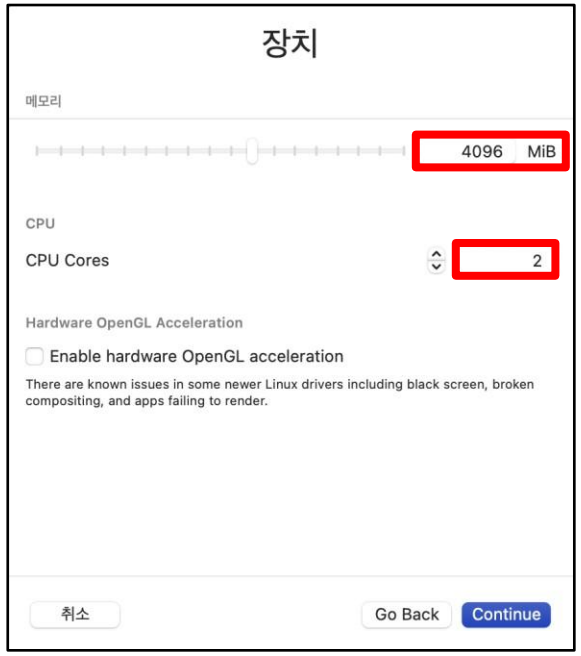
2)에서 다운받은 iso 이미지

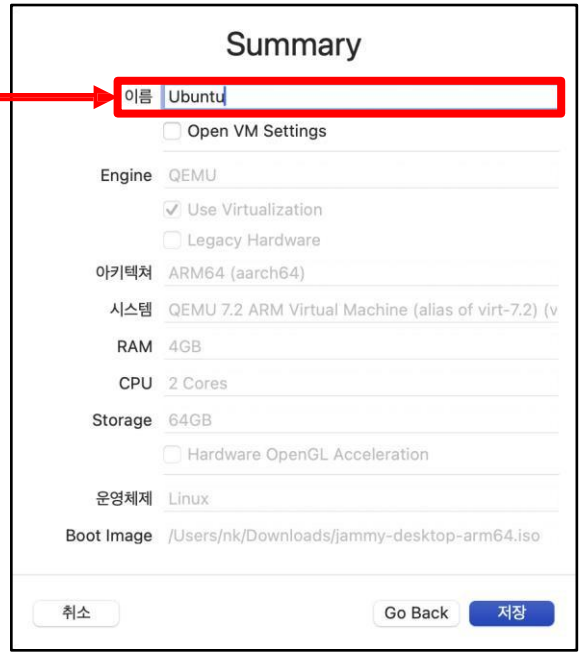
2. Mac (ARM)

3) VM 생성

- 메모리: 4096MB, Core: 2개, 디스크: 64GB
- 본인 컴퓨터 사양에 맞춰 설정 가능

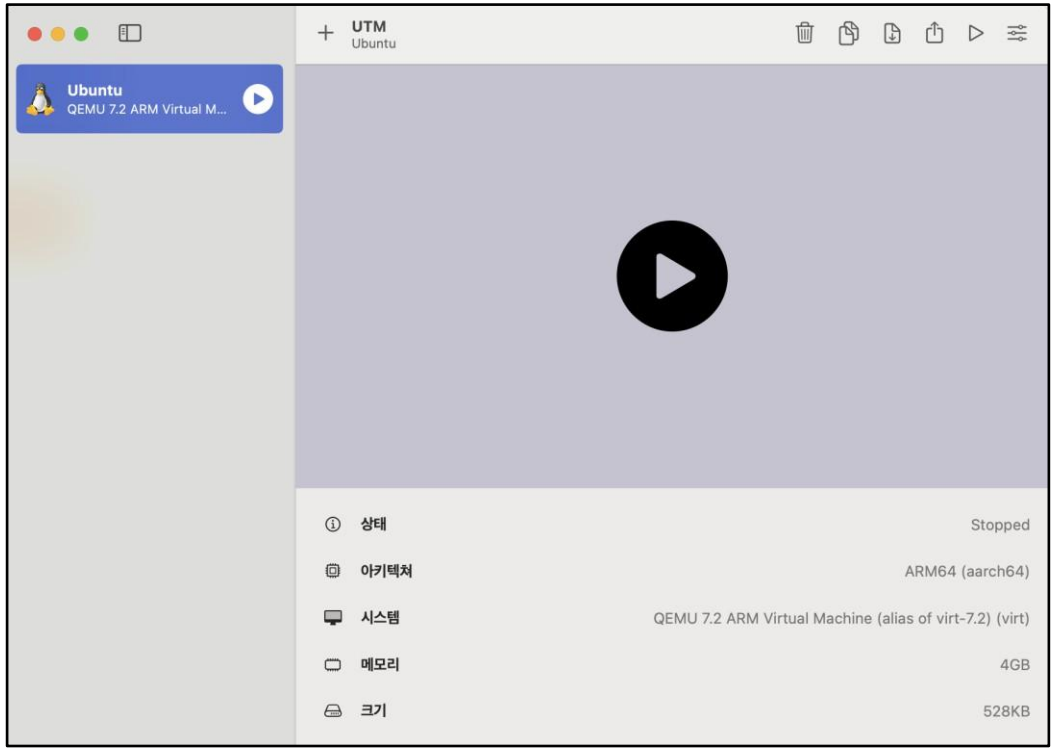
본인이 식별하기 쉬운 이름으로 설정





2. Mac (ARM)

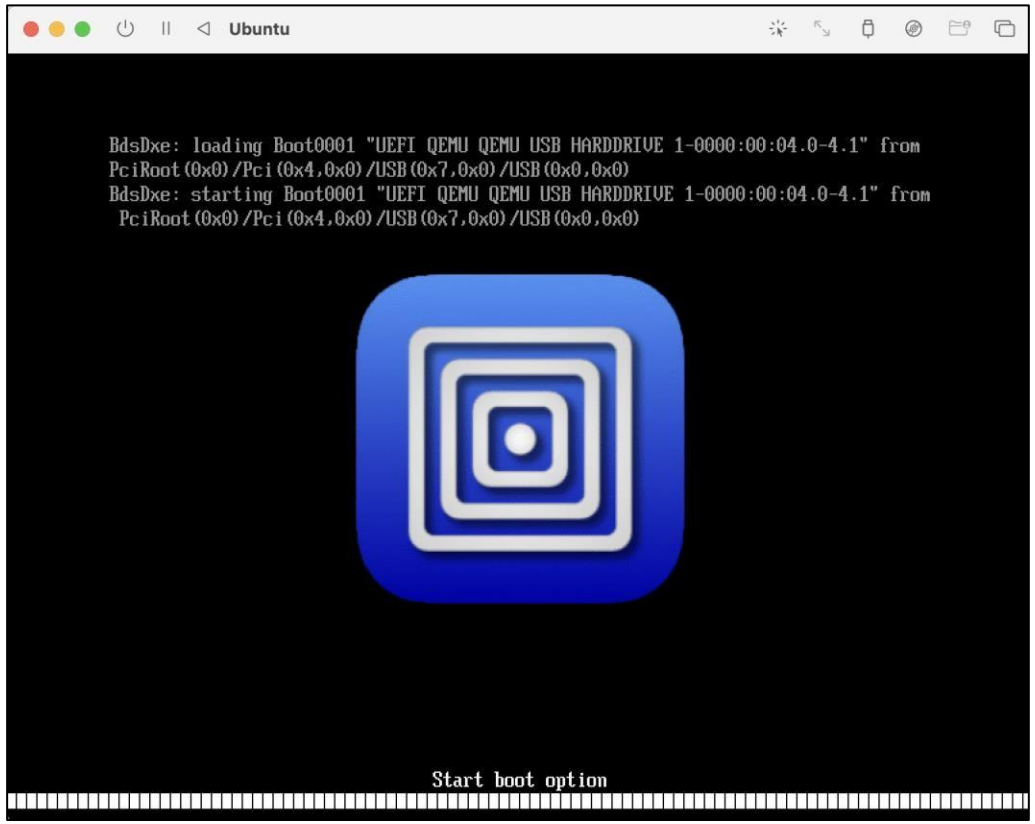
3) VM 생성 - 완료



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

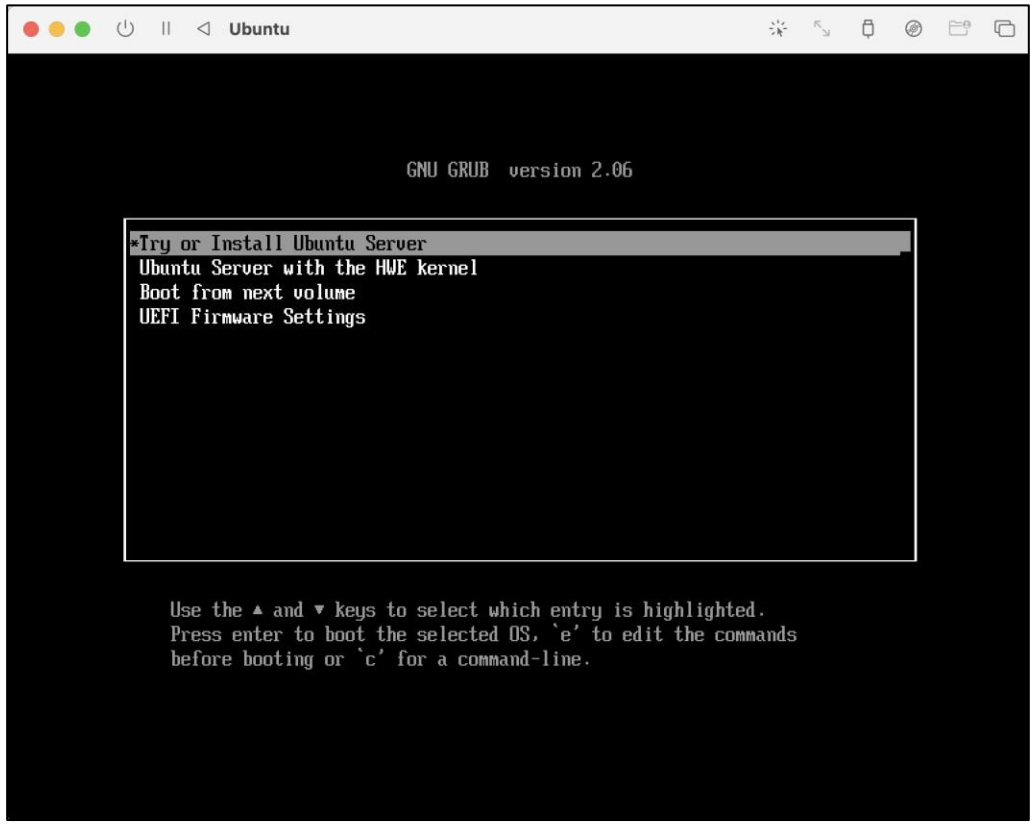
- 초기 VM 실행 화면



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

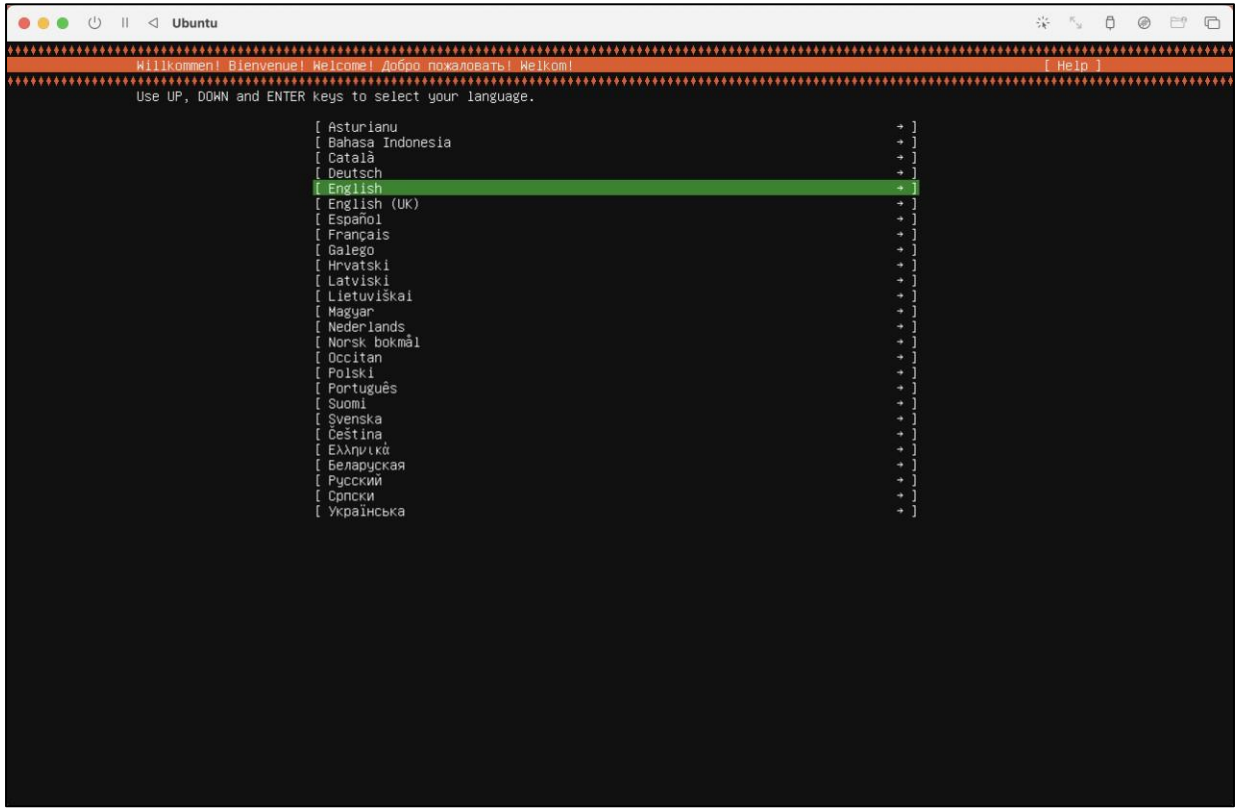
- 단계에 따라 설치 진행



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

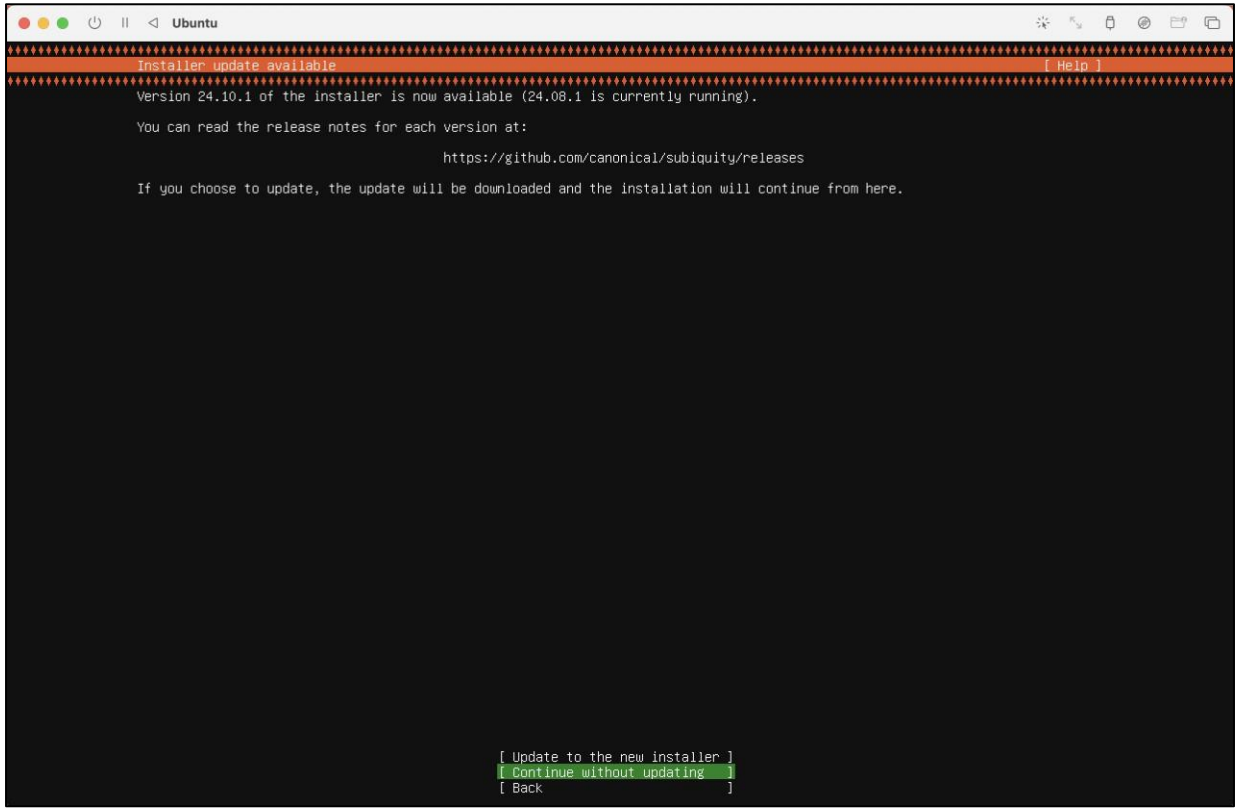
- 언어 설정: English



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

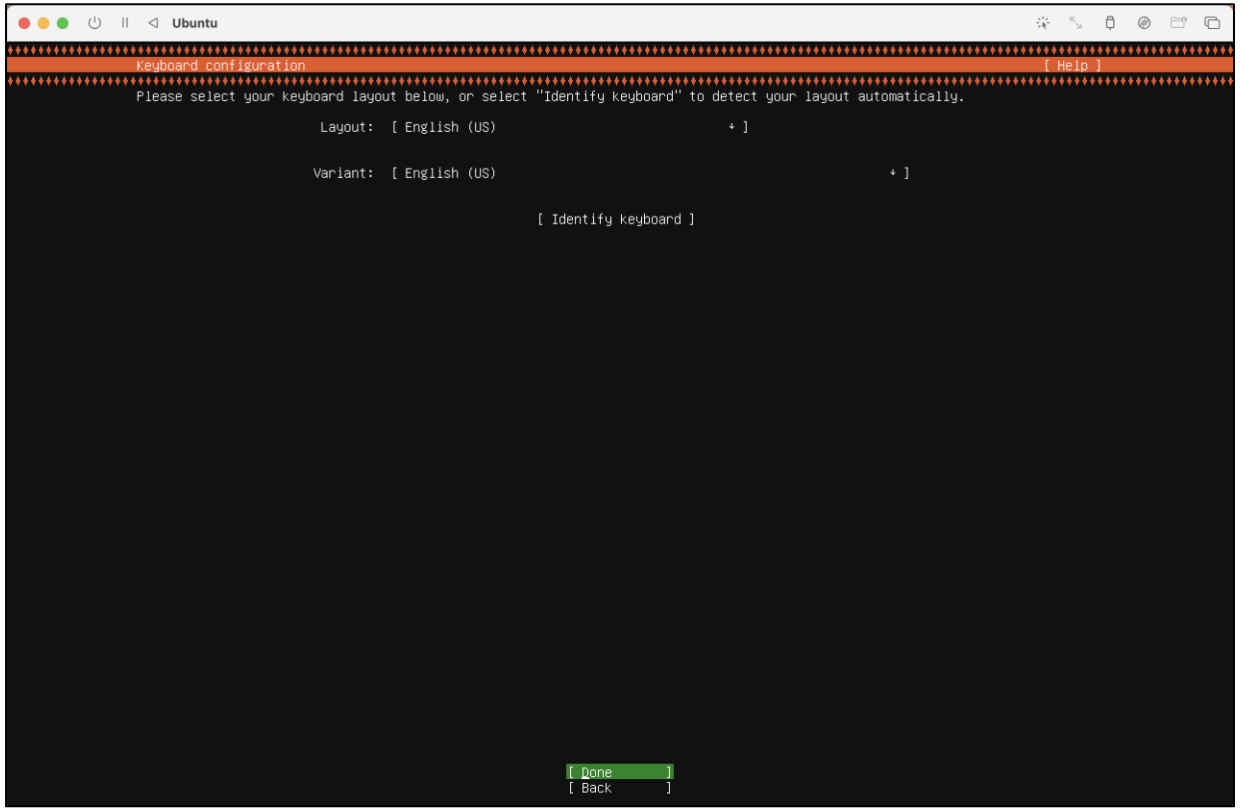
- 24버전으로 업데이트: 안함



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

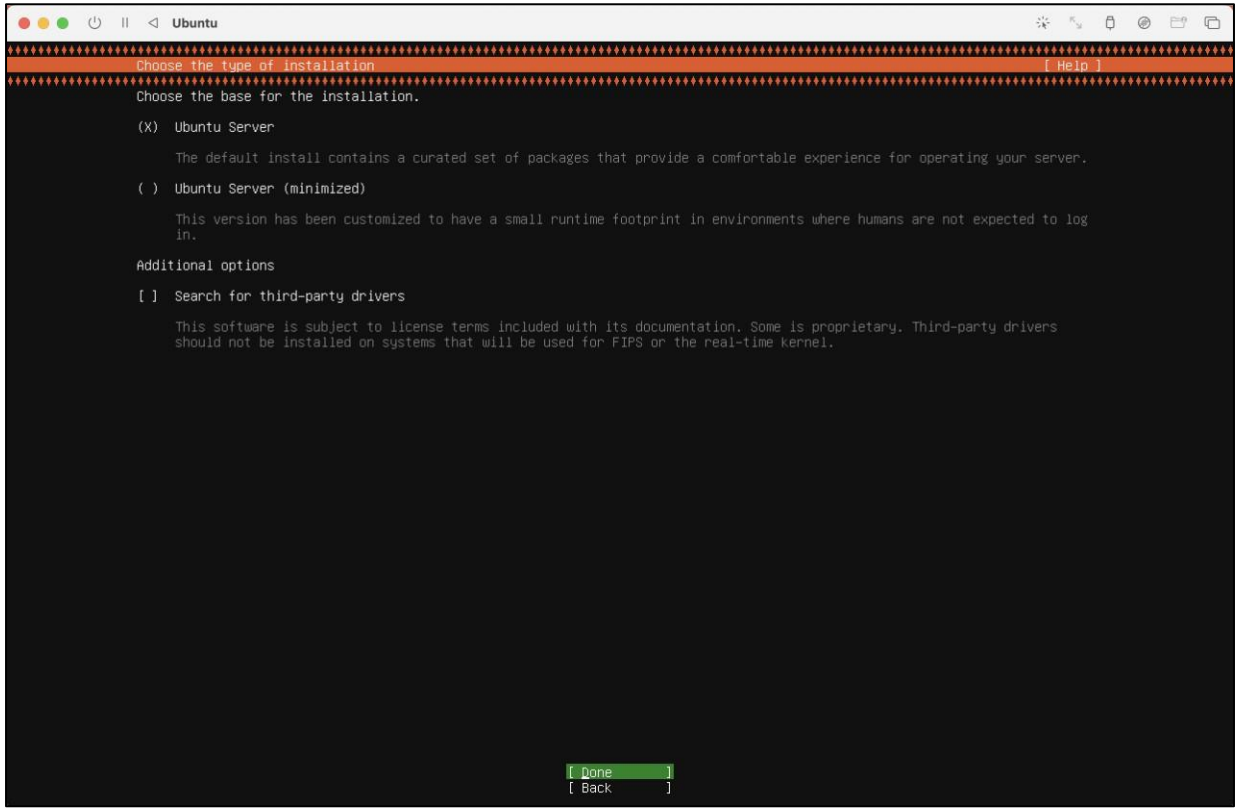
- 키보드 레이아웃: English



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

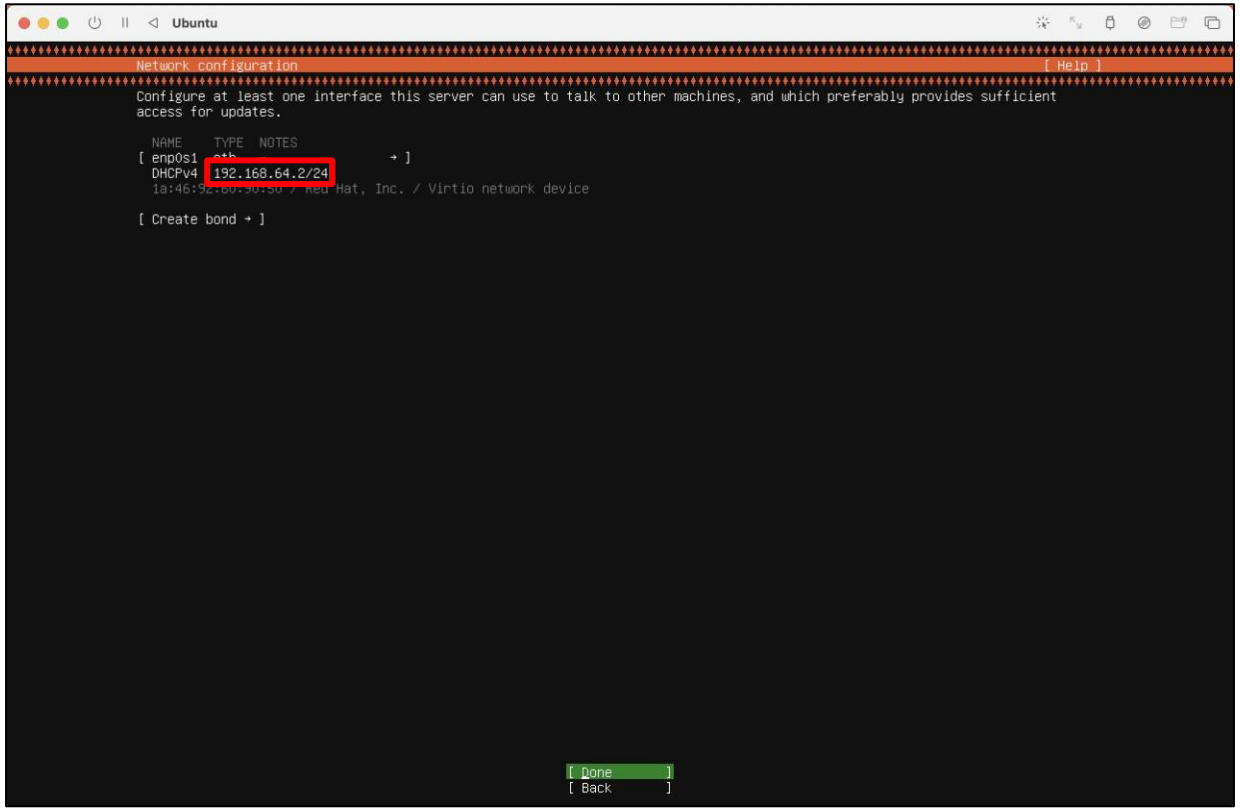
- Ubuntu 설치: Default



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

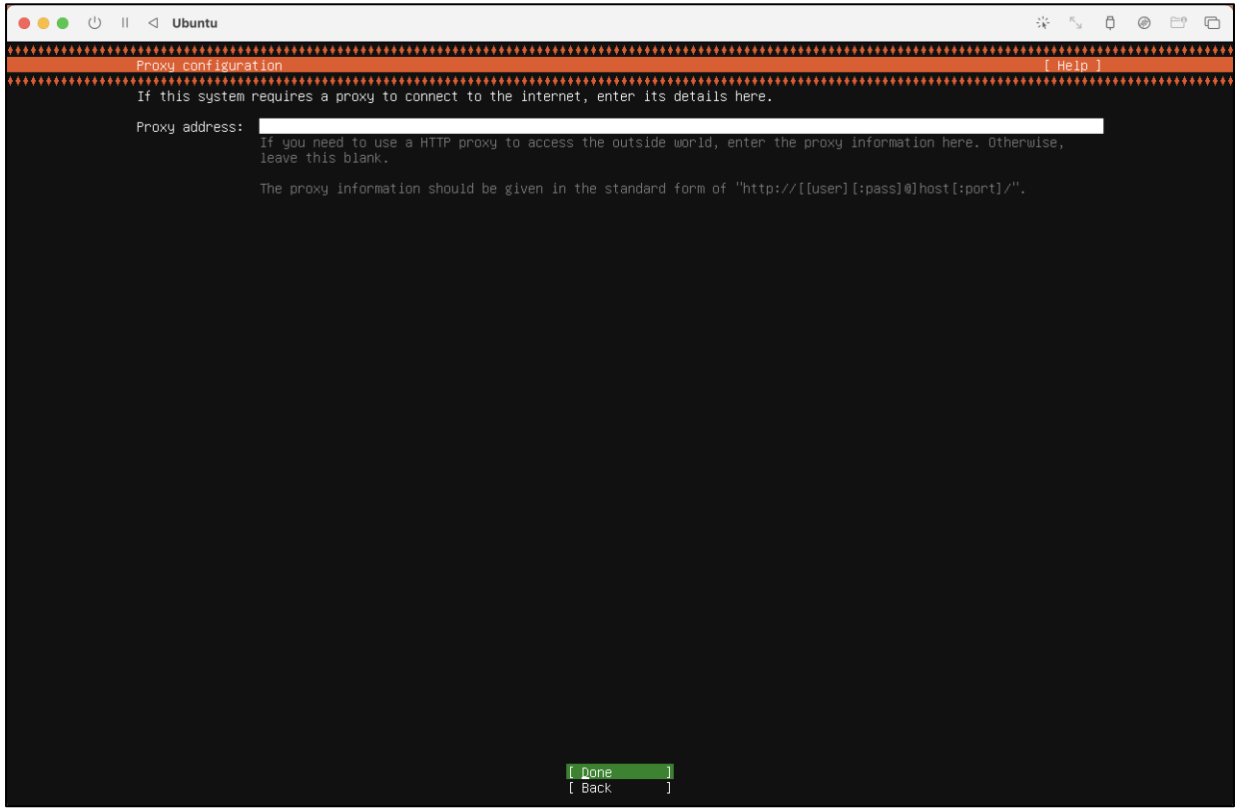
- 네트워크 설정: 그대로 진행, SSH를 사용할 것이라면 IP 주소 기억할 것



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

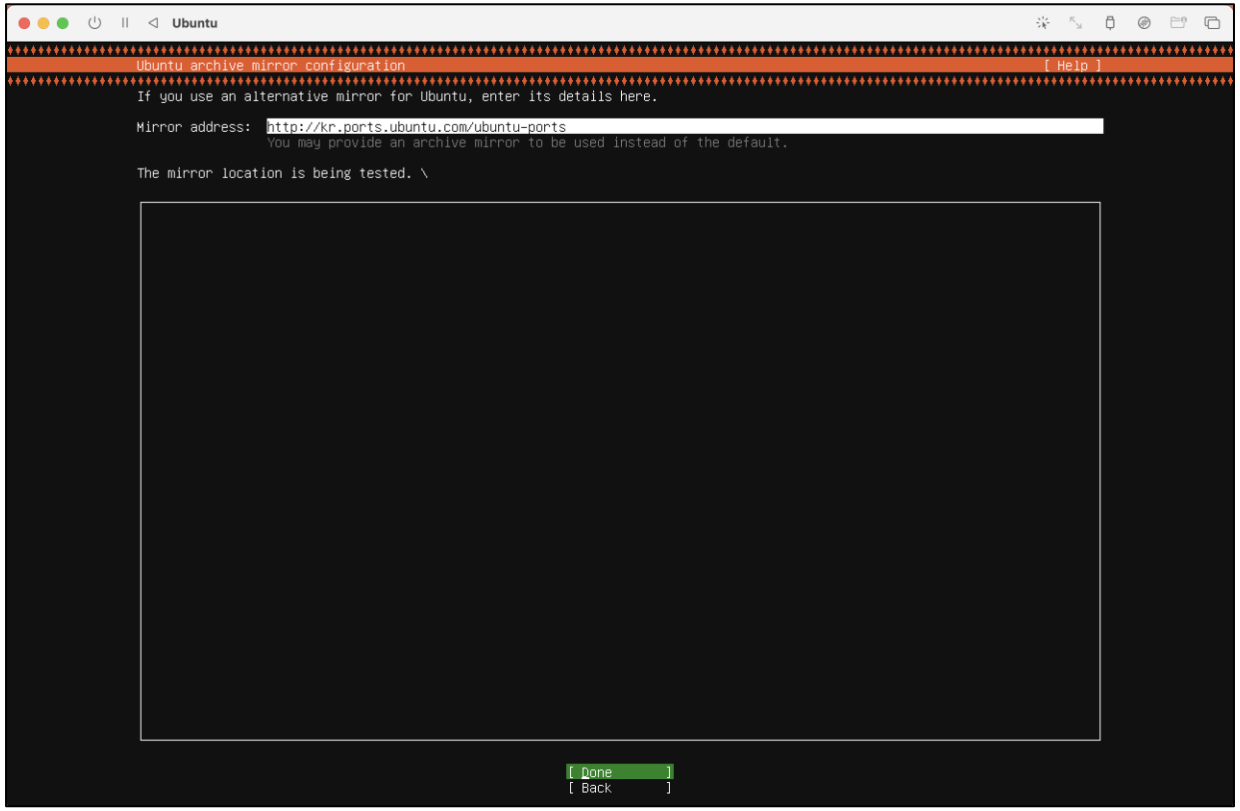
- Proxy 주소: 그대로 진행



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

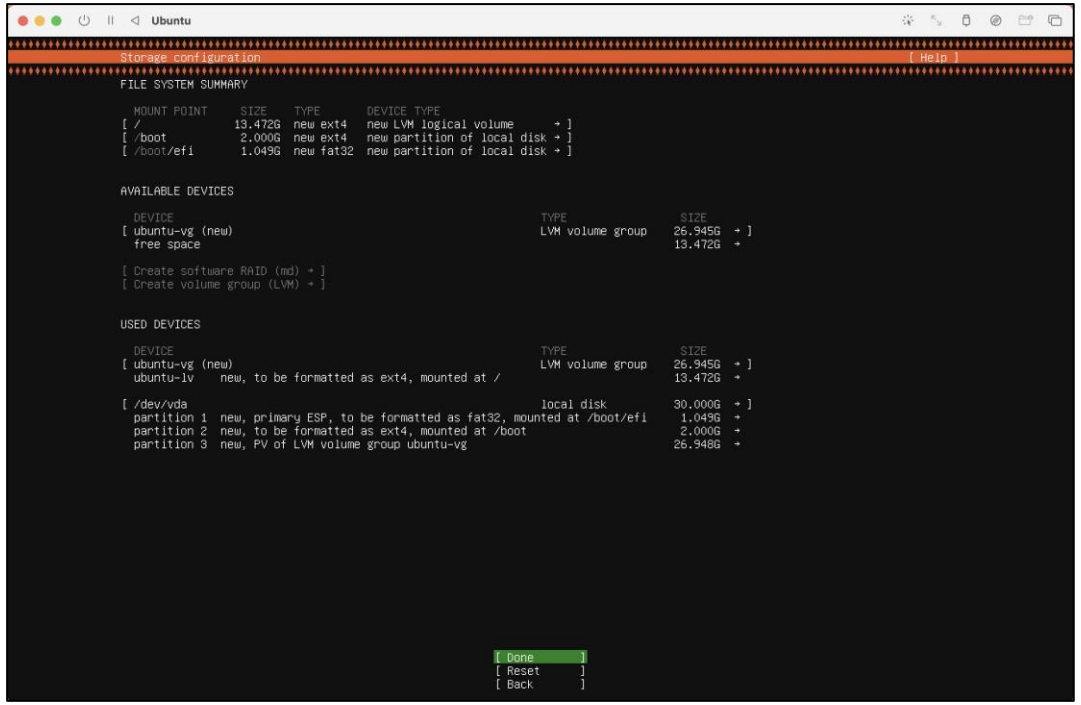
- Mirror 서버: 그대로 진행

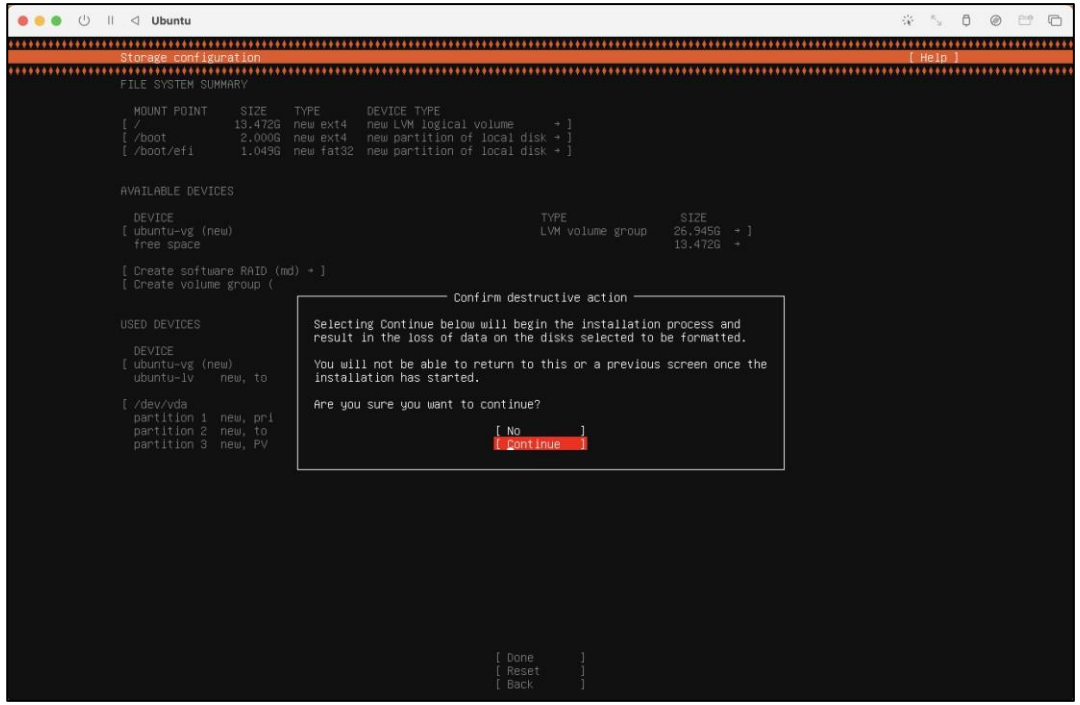


2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

- 파일 시스템 설정: 그대로 진행

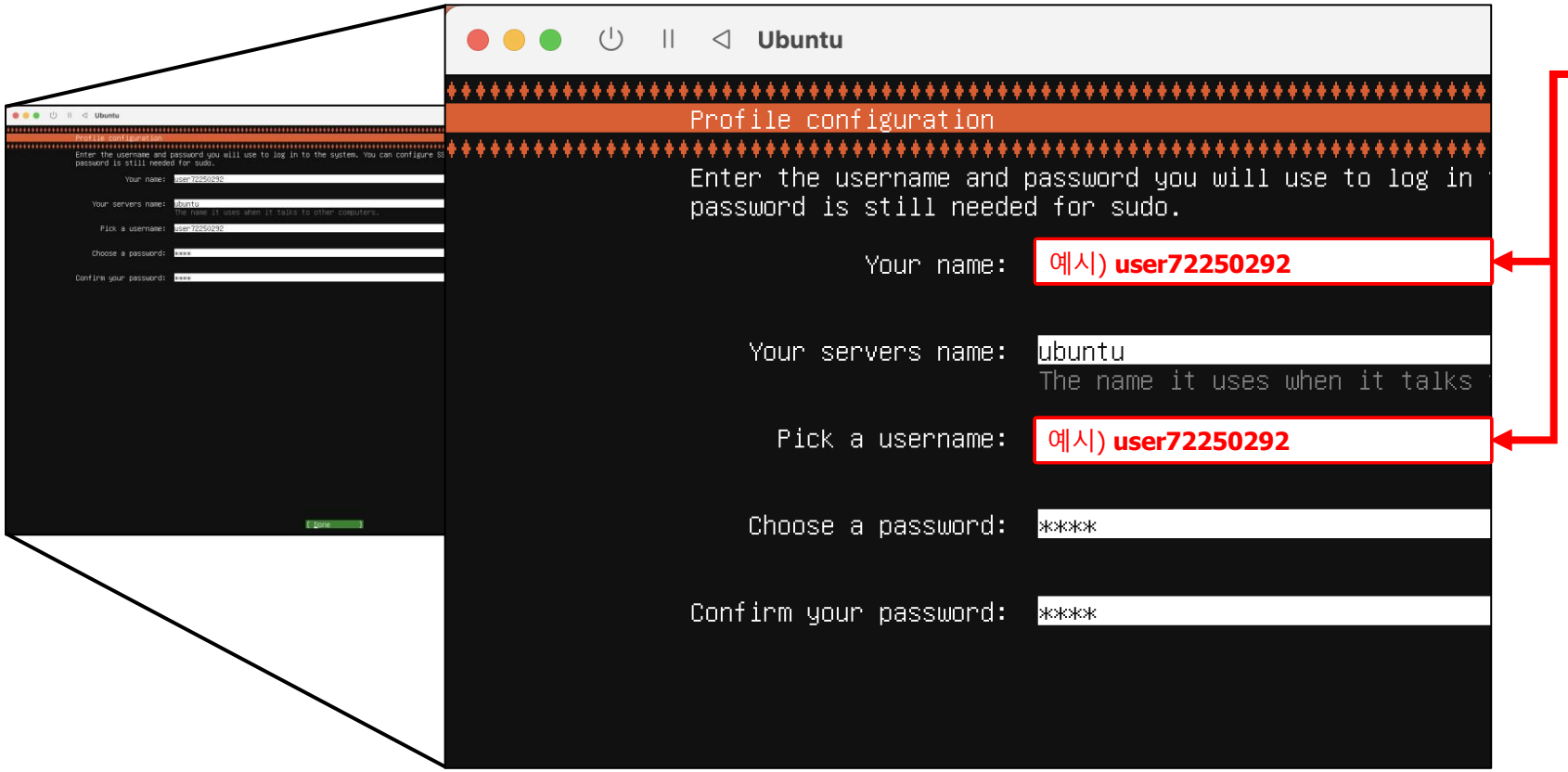




2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

- 계정 설정

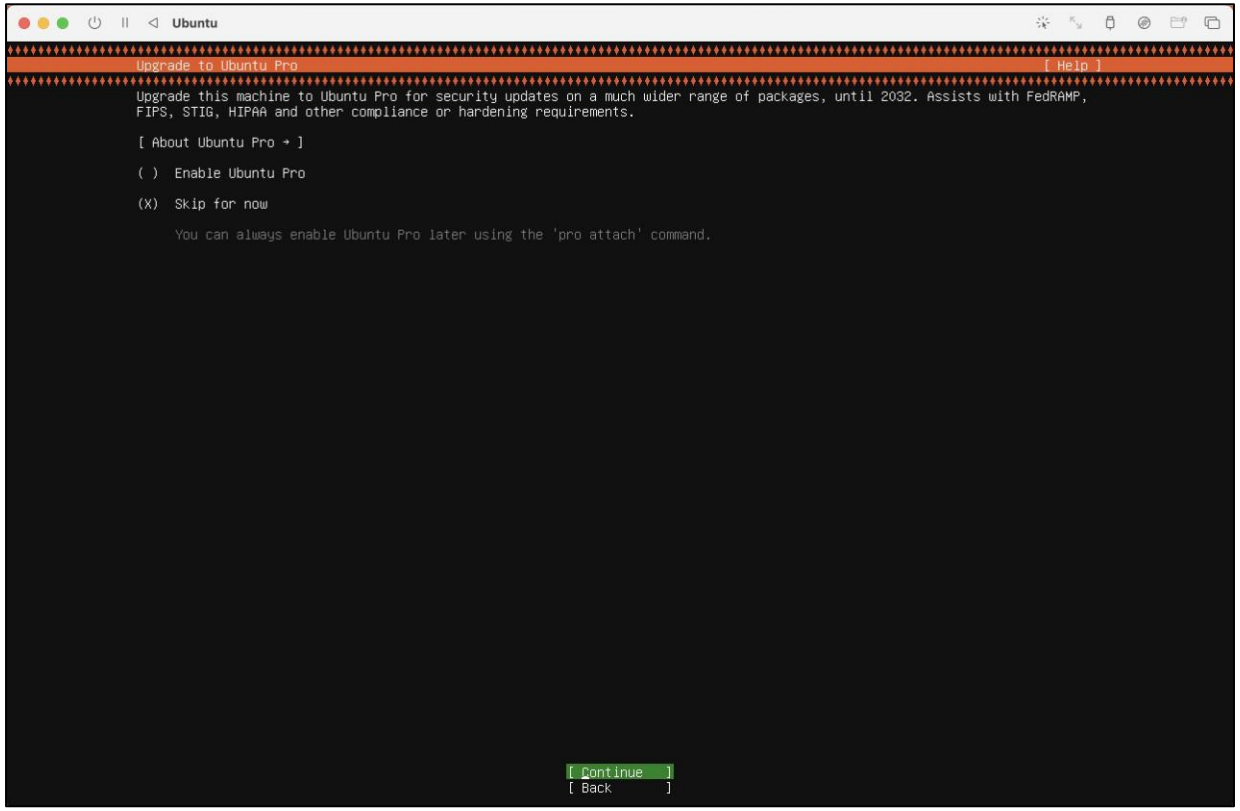


user + 학번으로 설정

2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

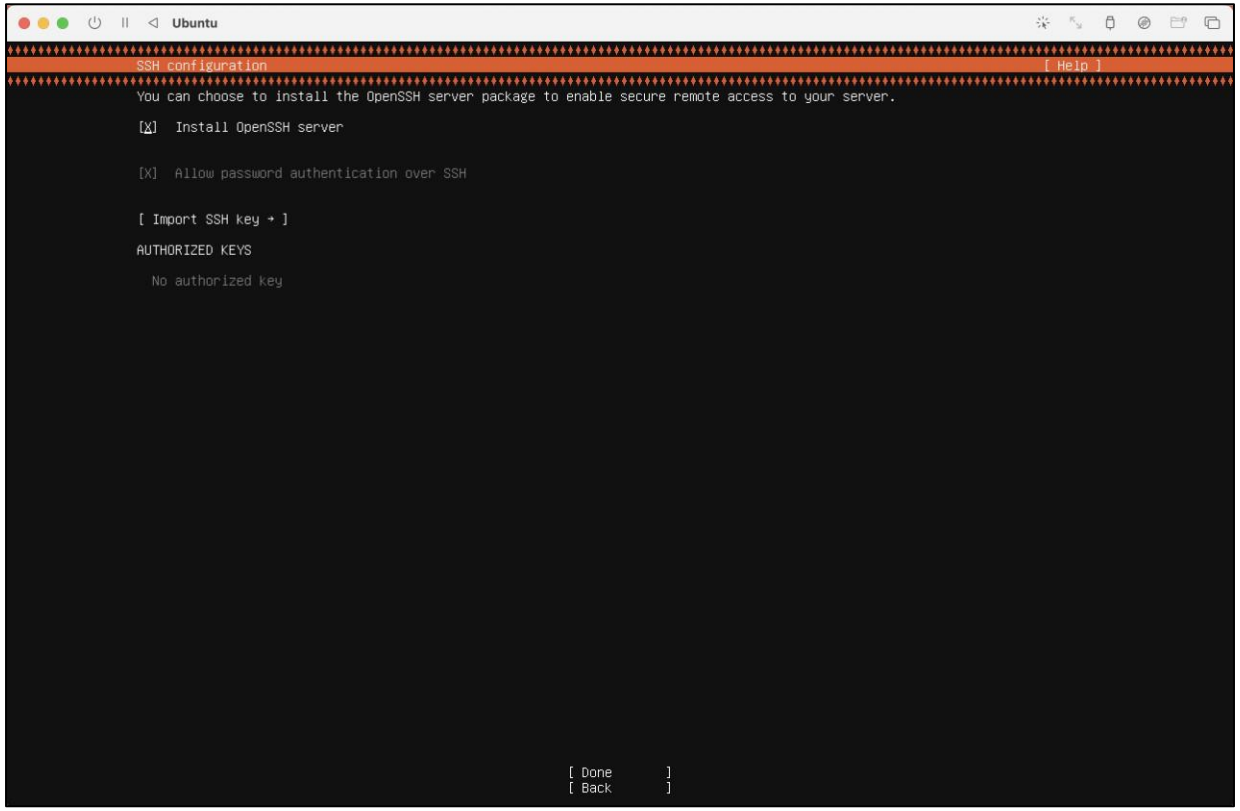
- Ubuntu pro로 업데이트: Skip



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

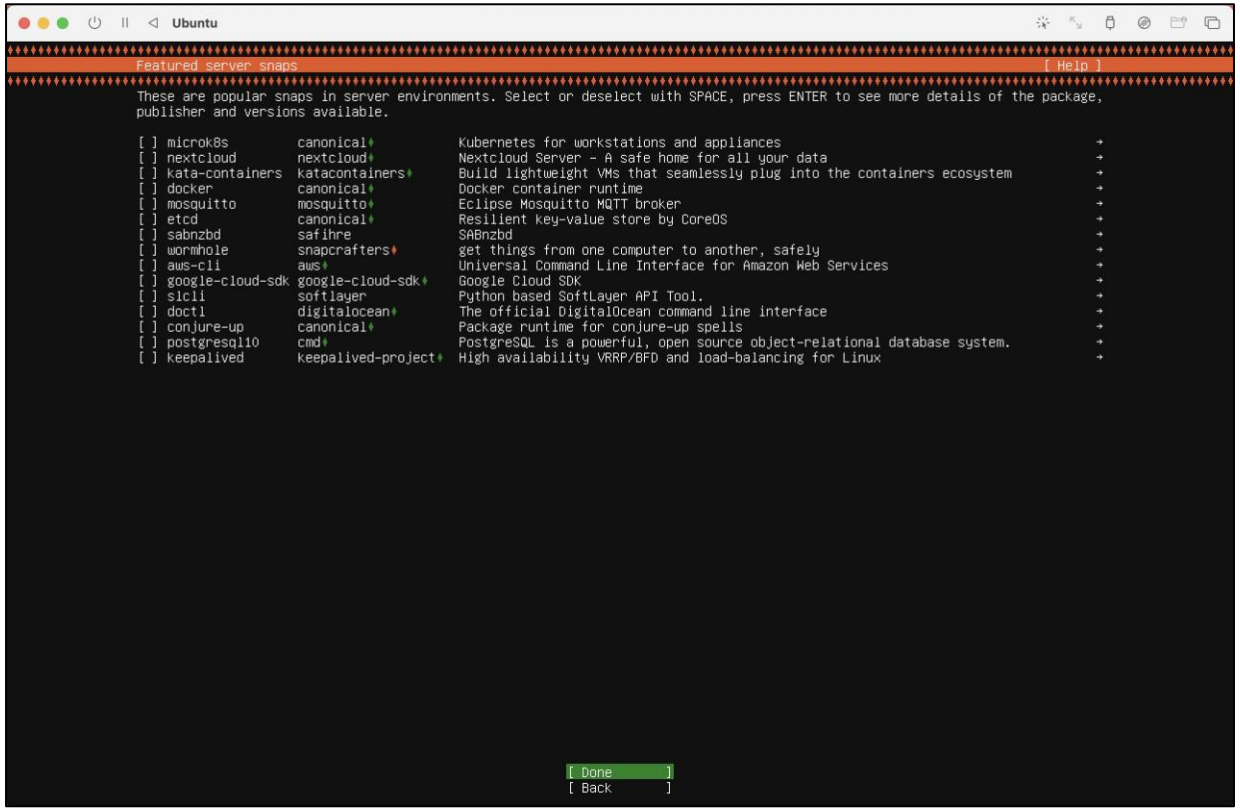
- SSH 서버 설정: SSH를 사용할 것이라면 체크 아니면 Skip



2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

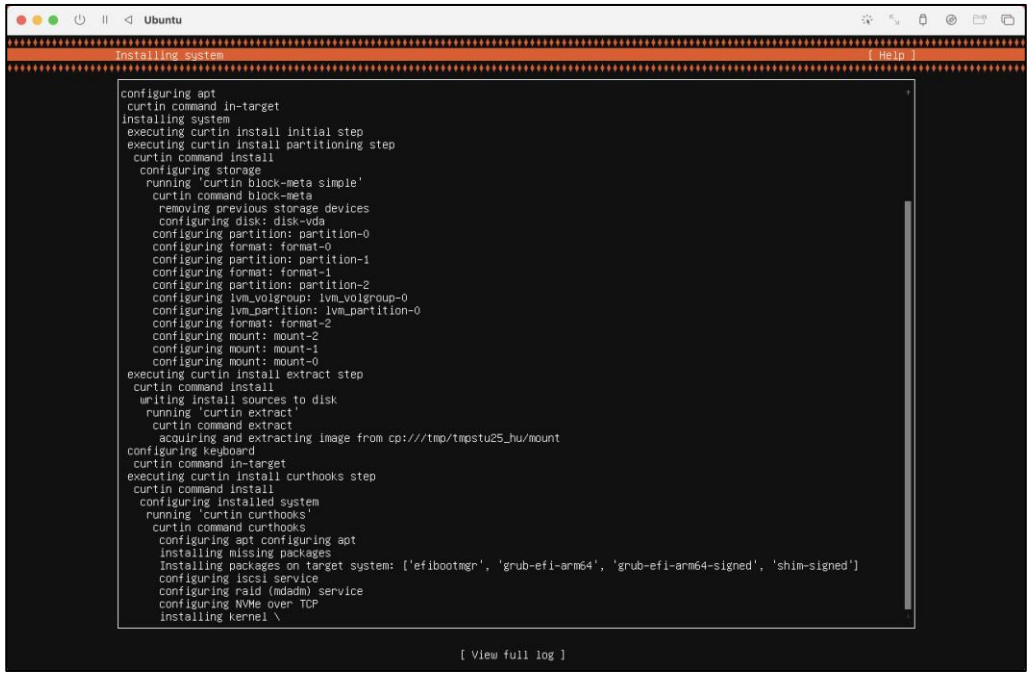
- 서버 snap 패키지 설치: Skip

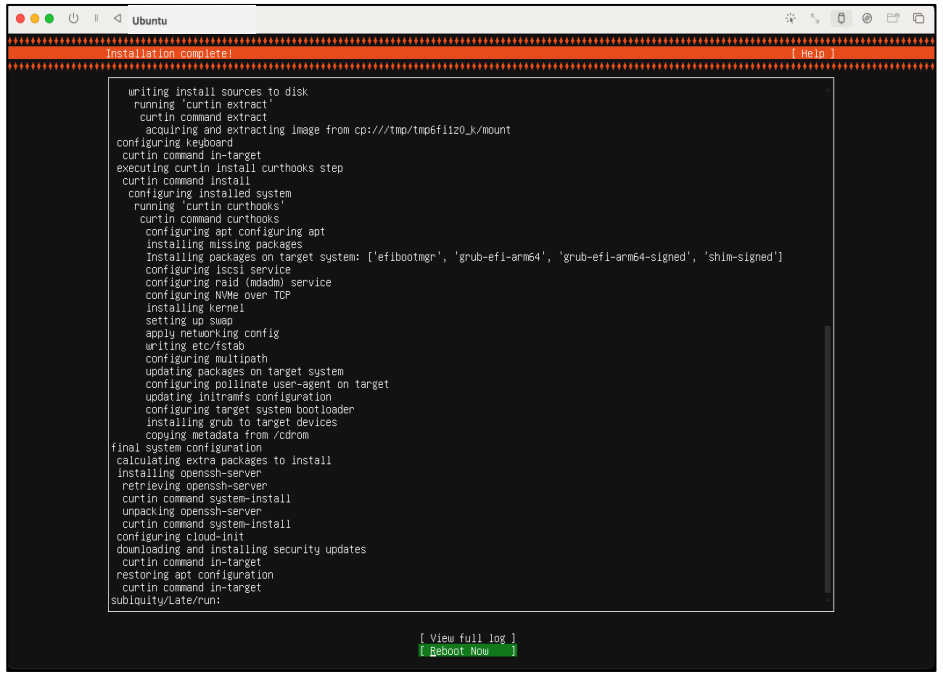


2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

- Ubuntu 설치 진행 화면, 완료되면 reboot

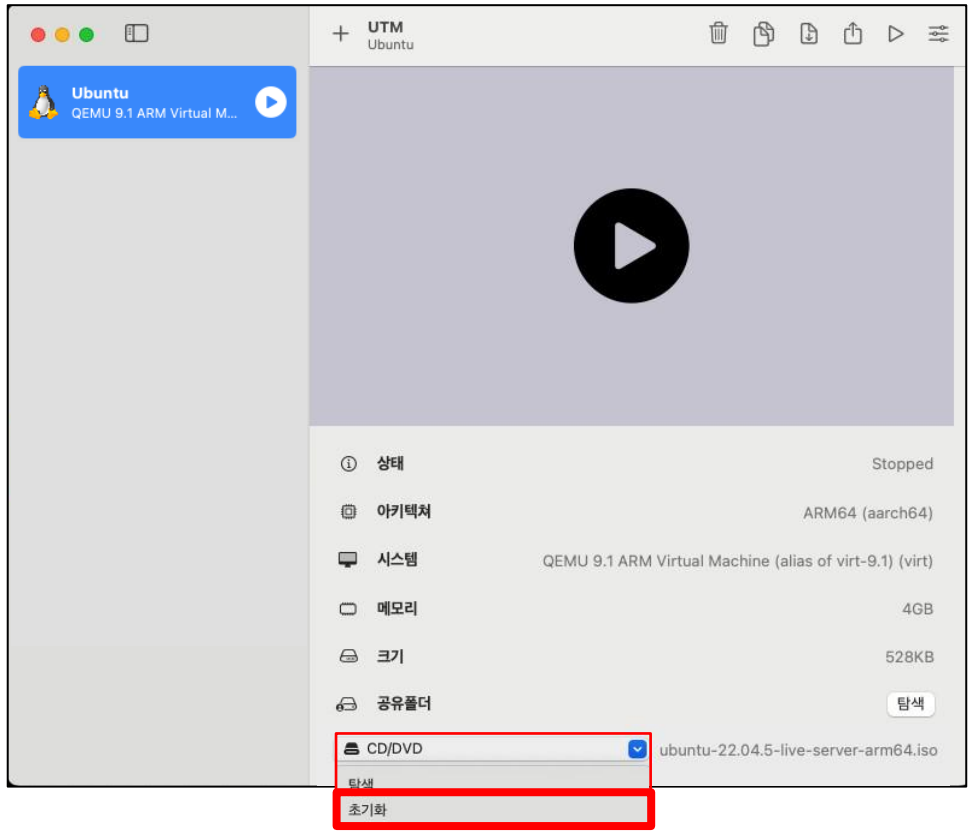




2. Mac (ARM)

4) 초기 설정 – Ubuntu 설치

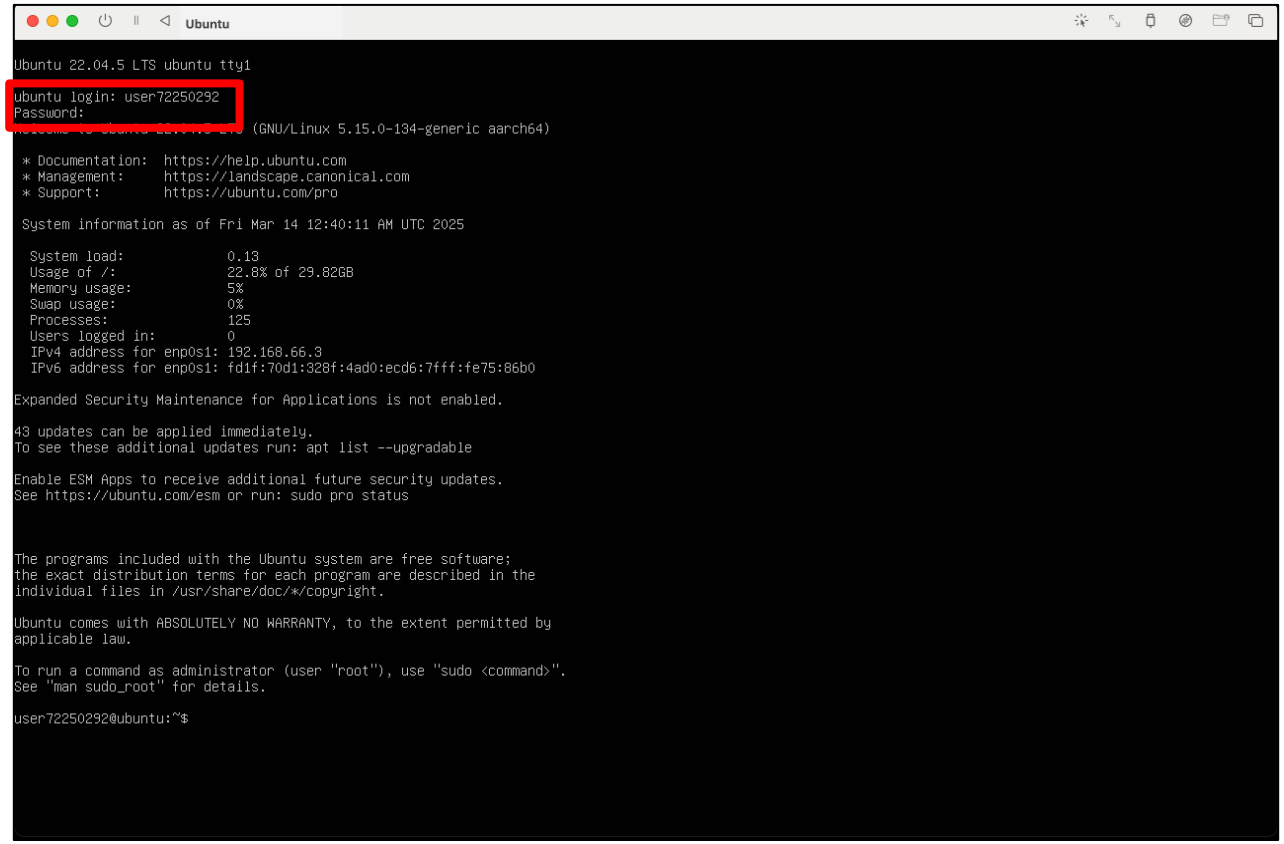
- 검은 화면이 뜬다면 VM 종료, CD/DVD 초기화 후 다시 시작



2. Mac (ARM)

4) 초기 설정 – 로그인

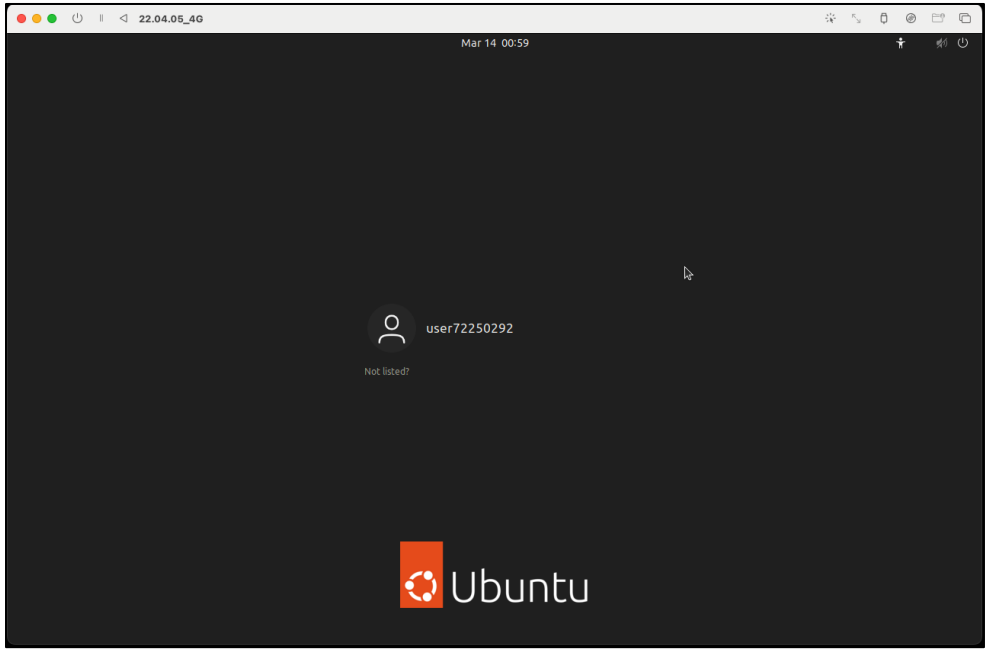
- {username} + 비밀번호 입력



2. Mac (ARM)

5) Optional – GUI 환경으로 변경

- $ sudo apt install ubuntu-desktop
- 설치 완료 후 $ sudo reboot
- reboot 후 다음 화면이 나온다면 성공



2. Mac (ARM)

5) Optional – SSH 설정

- Host(macOS)에서 터미널 실행 (VM은 실행 중인 상태)
- ~ ssh {username}@{IP address}
  - 39p에서 알아낸 IP 주소 또는 $ ip a로 알아낸 IP 주소

```
user72240257@ubuntu: ~                                    ⌥⌘1
→  ~ ssh user72240257@192.168.67.11
The authenticity of host '192.168.67.11 (192.168.67.11)' can't be established.
ED25519 key fingerprint is SHA256:ipfR5BZ9uBzoU1CmMiGDnv3eml011mxQ0O9t650o7tU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.67.11' (ED25519) to the list of known hosts.
user72240257@192.168.67.11's password:
Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 6.8.0-44-generic aarch64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro


Last login: Fri Sep  6 14:23:49 2024 from 192.168.67.1
user72240257@ubuntu:~$
```

Q&A?

2026.03.06

T.A. 위다연

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